Supplementary Material

Self-oscillating synchronematic colloids

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Supplementary Note 1: Analysis of oscillatory particle trajectories

Given tracking data for $k = 1, \ldots, N$ particle positions $\mathbf{y}_k$ at regular time intervals $t_k = k\Delta t$, we seek to determine latent (hidden) variables $\mathbf{s}_k$ that characterize the particle oscillator—namely, the oscillation phase θ_k, orientation angle χ_k, and position $\mathbf{x}_k$. To do so, we propose a generative, probabilistic model for the dynamics of an individual oscillator characterized by model parameters such as the oscillation frequency ω, phase diffusivity D_χ, angle diffusivity D_χ, etc. We use Bayesian inference to estimate both the latent variables and the model parameters from particle tracking data (Fig. S1). Below, we describe the model and outline the data analysis process; further details can be found in a forthcoming publication on characterizing "noise" in Quicke oscillators.

Generative Model

In the absence of noise, the transient position $\mathbf{y}(t)$ of an oscillating particle can be parameterized by the oscillator position $\mathbf{x}$, orientation χ, and phase $\theta(t)$ as

$$\mathbf{y}(t) = \mathbf{x} + \mathbf{n}(\chi) g(\theta(t)) \tag{1.1}$$

where $\mathbf{n}(\chi) = [\cos\chi, \sin\chi]^\top$ is a unit vector parallel to the oscillation direction, and $g(\theta)$ is the 2π-periodic waveform. By definition, the oscillation phase increases linearly with time as $\theta(t) = \omega t + \phi$, where ω is the oscillation frequency, and ϕ is the relative phase. Quincke oscillators are further distinguished by a symmetry whereby their dynamics is invariant to a simultaneous shift in phase $\theta \to \theta + \pi$ and angle $\chi \to \chi + \pi$.[1] This phase-angle symmetry implies the following constraint on the waveform: $g(\theta) = -g(\theta + \pi)$. We can therefore parameterize $g(\theta)$ by the Fourier series

$$g(\theta) = \sum_{n=1}^{\infty} b_n \cos[(2n-1)\theta] + c_n \sin[(2n-1)\theta] \tag{1.2}$$

where b_n and c_n are real-valued coefficients, collectively denoted $\mathbf{b}$. Without loss of generality, we set $c_1 = 0$, which determines the oscillation amplitude at zero phase, $\theta = 0$.

The noise-free oscillator is characterized by a constant position $\mathbf{x}$, orientation χ, and relative phase ϕ. In experiment, however, these variables are observed to change on time scales longer than the oscillation period $T = 2\pi/\omega$. We approximate these changes by a stochastic diffusion process parameterized by diffusion coefficients D_θ, D_χ, and D_x for the oscillator phase, angle, and position, respectively. Finally, to account for remaining discrepancies between the observed particle position and the predictions of equation (1.1), we model the measurement error by white Gaussian noise with zero mean and variance σ^2.

We denote the latent variables at time t_k by the state vector $\mathbf{s}_k = [\theta_k, \chi_k, x_{1k}, x_{2k}]^\top$. The joint distribution for the particle positions $\mathbf{y}_{1:N}$ and the latent variables $\mathbf{s}_{1:N}$ given the model parameters $\lambda = \{\omega, D_\theta, D_\chi, D_x, \mathbf{s}_0, \mathbf{b}, \sigma^2\}$ is

$$p(\mathbf{s}_{1:N}, \mathbf{y}_{1:N} \mid \lambda) = \prod_{k=1}^{N} p(\mathbf{s}_k \mid \mathbf{s}_{k-1}, \lambda) p(\mathbf{y}_k \mid \mathbf{s}_k, \lambda) \tag{1.3}$$

Here, the *process* model $p(\mathbf{s}_k \mid \mathbf{s}_{k-1}, \lambda)$ is a normal distribution with mean vector $\mathbf{s}_{k-1} + \mathbf{w}$ where $\mathbf{w} = [\omega\Delta t, 0, 0, 0]^\top$ and covariance matrix R_k given by

$$R_k = 2\Delta t \begin{bmatrix} D_\theta & 0 & 0 & 0 \\ 0 & D_\chi & 0 & 0 \\ 0 & 0 & D_x & 0 \\ 0 & 0 & 0 & D_x \end{bmatrix} \tag{1.4}$$

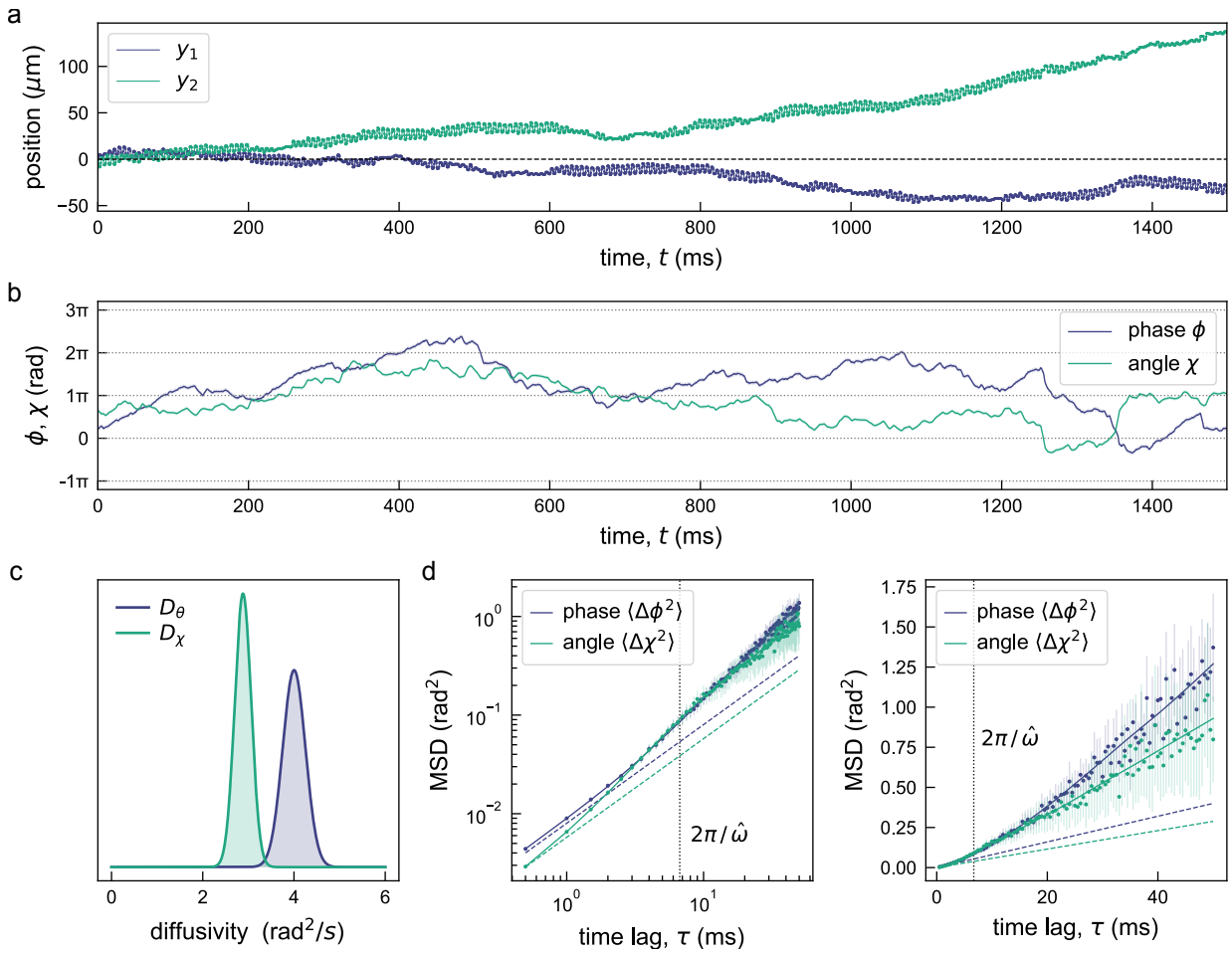

Figure S1: Analysis of an oscillatory particle trajectory. (a) Particle position $\mathbf{y}$ vs. time t at regular intervals of 0.5 ms from particle tracking (markers). The solid curve denotes the fitted trajectory based on the stochastic model described in the text. Particle radius is $a = 5$ μm; the AOT concentration is 150 mM; the external field strength is $E_e = 6$ V/μm. (b) Inferred time-series of the relative phase $\phi = \theta - \hat{\omega}t$ and angle χ of the particle oscillator. The inferred frequency is $\omega = 938 \pm 6$ rad/s. (c) Approximate posterior distributions for the phase diffusivity D_θ and angle diffusivity D_χ. (d) Mean squared displacement (MSD) of the relative phase ϕ and angle χ as a function of time lag τ. Dashed lines denote the diffusive expectation $2D\tau$ for the inferred parameters $D_\theta = 4.0 \pm 0.25$ rad^2/s and $D_\chi = 2.9 \pm 0.18$ rad^2/s. Similar analysis of additional particles suggests an average frequency of $\omega = 1060 \pm 23$ rad/s, an average phase diffusivity of $D_\theta = 5.2 \pm 1.2$ rad^2/s, and an average angle diffusivity of $D_\chi = 4.3 \pm 0.6$ rad^2/s.

The oscillator phase, angle, and position change in time due to diffusion with coefficients D_θ, D_χ, and D_x. Furthermore, the phase θ_k increases with a constant drift velocity ω equal to the oscillation frequency. The *measurement* model $p(\mathbf{y}_k \mid \mathbf{s}_k, \lambda)$ is also normally distributed with mean vector

$$\mathbf{h}(\mathbf{s}_k) = \mathbf{x}_k + \mathbf{n}(\chi_k) g(\theta_k) \tag{1.5}$$

and covariance matrix $\sigma^2 I$, where I is the 2×2 identity matrix.

Parameter Estimation: Expectation-Maximization Algorithm

The posterior distribution for the latent variables $\mathbf{s}_{1:N}$ and the model parameters λ given the observed positions $\mathbf{y}_{1:N}$ is

$$p(\mathbf{s}_{1:N}, \lambda \mid \mathbf{y}_{1:N}) \propto p(\mathbf{s}_{1:N}, \mathbf{y}_{1:N} \mid \lambda) p(\lambda) \tag{1.6}$$

For simplicity, we assume a uniform prior for the parameters such that $p(\lambda) =$ constant over the range of interest. The distribution $p(\mathbf{s}_{1:N}, \mathbf{y}_{1:N} \mid \lambda)$ is described by equation (1.3). We use the expectation maximization algorithm[2] to identify the MAP estimates of the latent variables $\hat{\mathbf{s}}_{1:N}$ and the model parameters $\hat{\lambda}$ that maximize the posterior distribution of equation (1.6). In the j^{th} step of this iterative process, we evaluate the distribution $p(\mathbf{s}_{1:N} \mid \mathbf{y}_{1:N}, \lambda_j)$ for the latent variables conditioned on the observed positions and the current estimate of the model parameters. Using this distribution, we compute the expectation of the log likelihood $\ln p(\mathbf{s}_{1:N}, \mathbf{y}_{1:N} \mid \lambda)$ and maximize this function with respect to λ to obtain an improved estimate λ_{j+1}. Repeated iteration of the expectation and maximization steps converges to the desired MAP estimates.

Expectation Step: Kalman Filtering & Smoothing

We approximate the distribution $p(\mathbf{s}_{1:N} \mid \mathbf{y}_{1:N}, \lambda)$ using an extended Kalman filter with Rauch-Tung-Striebel smoothing.[2–4] This algorithm provides an efficient approach that scales linearly with the number of time steps N.

Forward Kalman Filter. During the first "filter" step, we progress through the unobserved states $\mathbf{s}_k$ from $k = 1$ to N conditioning on all observations up to and including $\mathbf{y}_k$. We denote the resulting distribution for $\mathbf{s}_k$ as $p(\mathbf{s}_k \mid \mathbf{y}_{1:k})$, where $\mathbf{y}_{1:k} = \{\mathbf{y}_1, \ldots, \mathbf{y}_k\}$. To streamline the notation, we have omitted the conditioning on the model parameters λ throughout this section. In progressing from index $k-1$ to k, we apply the following Bayesian updating procedure

$$p(\mathbf{s}_k \mid \mathbf{y}_{1:k}) \propto p(\mathbf{y}_k \mid \mathbf{s}_k) p(\mathbf{s}_k \mid \mathbf{s}_{k-1}) p(\mathbf{s}_{k-1} \mid \mathbf{y}_{1:k-1}) \tag{1.7}$$

Here, the distributions $p(\mathbf{s}_k \mid \mathbf{y}_{1:k})$ are approximated by normal distributions with mean $\hat{\mathbf{s}}_{k|k}$ and variance $P_{k|k}$. Prior to observing the measurement $\mathbf{y}_k$, we can *predict* the unobserved state $\mathbf{s}_k$ conditioned on the prior observations $\mathbf{y}_{1:k-1}$ as

$$\begin{aligned} p(\mathbf{s}_k \mid \mathbf{y}_{1:k-1}) &\propto p(\mathbf{s}_k \mid \mathbf{s}_{k-1}) p(\mathbf{s}_{k-1} \mid \mathbf{y}_{1:k-1}) \\ &= \mathcal{N}(\mathbf{s}_k \mid \hat{\mathbf{s}}_{k|k-1}, P_{k|k-1}) \end{aligned} \tag{1.8}$$

with mean $\hat{\mathbf{s}}_{k|k-1}$ and covariance $P_{k|k-1}$ given by

$$\begin{aligned} \hat{\mathbf{s}}_{k|k-1} &= \hat{\mathbf{s}}_{k-1|k-1} + \mathbf{w} \\ P_{k|k-1} &= P_{k-1|k-1} + R_k \end{aligned} \tag{1.9}$$

where the process model covariance R_k is given by equation (1.4). Using this estimate for $\mathbf{s}_k$, we linearize the measurement model to obtain a tractable linear model of the form

$$\mathbf{h}_k(\mathbf{s}_k) \approx \mathbf{h}_k(\hat{\mathbf{s}}_{k|k-1}) + (\mathbf{s}_k - \hat{\mathbf{s}}_{k|k-1}) \cdot \nabla_{\mathbf{s}_k} \mathbf{h}(\hat{\mathbf{s}}_{k|k-1}) = H_k \mathbf{s}_k + \mathbf{v}_k \tag{1.10}$$

where the matrix H_k and and vector $\mathbf{v}_k$ are given by

$$\begin{aligned} H_k &= \begin{bmatrix} b\cos(\hat{\chi}_{k|k-1})g'(\hat{\theta}_{k|k-1}) & -b\sin(\hat{\chi}_{k|k-1})g(\hat{\theta}_{k|k-1}) & 1 & 0 \\ b\sin(\hat{\chi}_{k|k-1})g'(\hat{\theta}_{k|k-1}) & b\cos(\hat{\chi}_{k|k-1})g(\hat{\theta}_{k|k-1}) & 0 & 1 \end{bmatrix} \\ \mathbf{v}_k &= \mathbf{h}_k(\hat{\mathbf{s}}_{k|k-1}) - H_k \hat{\mathbf{s}}_{k|k-1} \end{aligned} \tag{1.11}$$

This approach based on linearizing the nonlinear model around the current state estimate is known as the extended Kalman filter (EKF). Using the linearized measurement model, we can *update* the prior estimate $\hat{\mathbf{s}}_{k|k-1}$ on the latest observation $\mathbf{y}_k$ as

$$\begin{aligned} p(\mathbf{s}_k \mid \mathbf{y}_{1:k}) &\propto p(\mathbf{y}_k \mid \mathbf{s}_k) p(\mathbf{s}_k \mid \mathbf{y}_{1:k-1}) \\ &= \mathcal{N}(\mathbf{s}_k \mid \hat{\mathbf{s}}_{k|k}, P_{k|k}) \end{aligned} \tag{1.12}$$

with mean $\hat{\mathbf{s}}_{k|k}$ and covariance $P_{k|k}$ given by

$$\begin{aligned} \hat{\mathbf{s}}_{k|k} &= P_{k|k}\big(P_{k|k-1}^{-1}\hat{\mathbf{s}}_{k|k-1} + \sigma^{-2}H_k^\top(\mathbf{y}_k - \mathbf{v}_k)\big) \\ P_{k|k} &= \big(P_{k|k-1}^{-1} + \sigma^{-2}H_k^\top H_k\big)^{-1} \end{aligned} \tag{1.13}$$

Backward Kalman Smoothing. During the second "smoothing" step, we progress backwards through the unobserved states $\mathbf{s}_k$ from $k = N$ to 1 conditioning on all observations $\mathbf{y}_{1:N}$. In progressing from index $k+1$ to k, we use the following relationship

$$p(\mathbf{s}_k \mid \mathbf{y}_{1:N}) = p(\mathbf{s}_k \mid \mathbf{y}_{1:k}) \int \frac{p(\mathbf{s}_{k+1} \mid \mathbf{s}_k) p(\mathbf{s}_{k+1} \mid \mathbf{y}_{1:N})}{p(\mathbf{s}_{k+1} \mid \mathbf{y}_{1:k})} d\mathbf{s}_{k+1} \tag{1.14}$$

which involves the following distributions defined above

$$\begin{aligned} p(\mathbf{s}_k \mid \mathbf{y}_{1:k}) &= \mathcal{N}(\mathbf{s}_k \mid \hat{\mathbf{s}}_{k|k}, P_{k|k}) \\ p(\mathbf{s}_{k+1} \mid \mathbf{s}_k) &= \mathcal{N}(\mathbf{s}_{k+1} \mid \mathbf{s}_k + \mathbf{w}, R_k) \\ p(\mathbf{s}_{k+1} \mid \mathbf{y}_{1:k}) &= \mathcal{N}(\mathbf{s}_{k+1} \mid \hat{\mathbf{s}}_{k+1|k}, P_{k+1|k}) \end{aligned} \tag{1.15}$$

Due to the linear process model and the linearized measurement model, the posterior distribution $p(\mathbf{s}_k \mid \mathbf{y}_{1:N})$ is also normally distributed with mean $\hat{\mathbf{s}}_{k|N}$ and covariance $P_{k|N}$. Using equation (1.14), we can related these quantities for the k time step to the those of the $k+1$ time step as

$$\begin{aligned} \hat{\mathbf{s}}_{k|N} &= \hat{\mathbf{s}}_{k|k} + C_k(\hat{\mathbf{s}}_{k+1|N} - \hat{\mathbf{s}}_{k+1|k}) \\ P_{k|N} &= P_{k|k} + C_k(P_{k+1|N} - P_{k+1|k})C_k^\top \end{aligned} \tag{1.16}$$

where $C_k = P_{k|k}P_{k+1|k}^{-1}$. The inference of hidden state variables $\mathbf{s}_{1:N}$ via Kalman filtering and smoothing depends on the ability of the linearized measurement model to accurately approximate the full nonlinear problem. This accuracy can be improved by iterative application of the the filtering and smoothing steps, in which the measurement model is linearized about the previous posterior estimates $\hat{\mathbf{s}}_{k|N}$.

Supplementary Note 2: Model dynamics of a single Quincke oscillator

2.1 Moment dynamics

We consider a spherical particle of radius a, permittivity ε_p, and conductivity κ_p immersed in a fluid of permittivity ε_f and conductivity $\kappa_\mathrm{f}(\mathbf{r})$. The particle is subject to an external potential $\varphi_\mathrm{e}(\mathbf{r})$ due both to an applied field and to disturbance fields caused by neighboring particles. As derived previously,[1] the particle charge q, dipole $\mathbf{p}$, and quadrupole $\mathbf{Q}$ moments evolve due to charge accumulation at the particle-fluid interface and particle rotation with angular velocity $\boldsymbol{\Omega}$

$$\frac{dq}{dt} = -\frac{1}{\tau}q - \frac{\boldsymbol{\gamma}}{\tau}\cdot\left(\frac{2}{3a}\mathbf{p} - \frac{a^2}{3}\nabla\varphi_\mathrm{e}(\mathbf{0})\right) \tag{2.17}$$

$$\begin{aligned}\frac{d\mathbf{p}}{dt} &= \boldsymbol{\Omega}\times\left(\mathbf{p} + a^3\varepsilon'_\mathrm{cm}\nabla\varphi_\mathrm{e}(\mathbf{0})\right) - \frac{1}{\tau'}\left(\mathbf{p} + a^3\kappa'_\mathrm{cm}\nabla\varphi_\mathrm{e}(\mathbf{0})\right)\\ &\quad - \frac{\boldsymbol{\gamma}'}{\tau'}\cdot\left(aq\boldsymbol{\delta} + \frac{3}{5a}\mathbf{Q} - \frac{2a^4}{5}\nabla\nabla\varphi_\mathrm{e}(\mathbf{0})\right)\end{aligned} \tag{2.18}$$

$$\begin{aligned}\frac{d\mathbf{Q}}{dt} &= \left[\boldsymbol{\Omega}\times\left(2\mathbf{Q} + 4a^5\varepsilon''_\mathrm{cm}\nabla\nabla\varphi_\mathrm{e}(\mathbf{0})\right)\right]^\mathrm{st} - \frac{1}{\tau''}\left(\mathbf{Q} + 2a^5\kappa''_\mathrm{cm}\nabla\nabla\varphi_\mathrm{e}(\mathbf{0})\right)\\ &\quad - \frac{1}{\tau''}\left[\boldsymbol{\gamma}''\left(4a\mathbf{p} - 2a^4\nabla\varphi_\mathrm{e}(\mathbf{0})\right)\right]^\mathrm{st}\end{aligned} \tag{2.19}$$

where the potential gradients are evaluated at the particle center $\mathbf{r} = 0$. The time scales for the relaxation of the charge, dipole, and quadrupole are, respectively,

$$\tau = \frac{\varepsilon_\mathrm{f}}{\kappa_\mathrm{f}(\mathbf{0})} \quad , \quad \tau' = \frac{\varepsilon_\mathrm{p} + 2\varepsilon_\mathrm{f}}{\kappa_\mathrm{p} + 2\kappa_\mathrm{f}(\mathbf{0})} \quad , \quad \tau'' = \frac{2\varepsilon_\mathrm{p} + 3\varepsilon_\mathrm{f}}{2\kappa_\mathrm{p} + 3\kappa_\mathrm{f}(\mathbf{0})} \tag{2.20}$$

Similarly, the dimensionless conductivity gradients are

$$\boldsymbol{\gamma} = \frac{a\nabla\kappa_\mathrm{f}(\mathbf{0})}{\kappa_\mathrm{f}(\mathbf{0})} \quad , \quad \boldsymbol{\gamma}' = \frac{a\nabla\kappa_\mathrm{f}(\mathbf{0})}{\kappa_\mathrm{p} + 2\kappa_\mathrm{f}(\mathbf{0})} \quad , \quad \boldsymbol{\gamma}'' = \frac{a\nabla\kappa_\mathrm{f}(\mathbf{0})}{2\kappa_\mathrm{p} + 3\kappa_\mathrm{f}(\mathbf{0})} \tag{2.21}$$

The Claussius-Mossotti factors for the dipole and quadrupole are

$$\begin{aligned}\varepsilon'_\mathrm{cm} &= \frac{\varepsilon_\mathrm{p} - \varepsilon_\mathrm{f}}{\varepsilon_\mathrm{p} + 2\varepsilon_\mathrm{f}} \quad , \quad \varepsilon''_\mathrm{cm} = \frac{\varepsilon_\mathrm{p} - \varepsilon_\mathrm{f}}{2\varepsilon_\mathrm{p} + 3\varepsilon_\mathrm{f}}\\ \kappa'_\mathrm{cm} &= \frac{\kappa_\mathrm{p} - \kappa_\mathrm{f}(\mathbf{0})}{\kappa_\mathrm{p} + 2\kappa_\mathrm{f}(\mathbf{0})} \quad , \quad \kappa''_\mathrm{cm} = \frac{\kappa_\mathrm{p} - \kappa_\mathrm{f}(\mathbf{0})}{2\kappa_\mathrm{p} + 3\kappa_\mathrm{f}(\mathbf{0})}\end{aligned} \tag{2.22}$$

To preserve the properties of the quadrupole, which is symmetric and traceless, the bracketed terms with superscript 'st' denote the symmetric and traceless components of the second order tensor—that is, $[\mathbf{A}]^\mathrm{st} = \frac{1}{2}(\mathbf{A} + \mathbf{A}^\intercal) - \frac{1}{3}\mathrm{tr}(\mathbf{A})\boldsymbol{\delta}$, where $\mathbf{A}^\intercal$ denotes the transpose, $\boldsymbol{\delta}$ is the identity tensor, and $\mathrm{tr}(\mathbf{A}) = A_{ii}$ denotes the trace. Equation (2.19) neglects the additional contributions due to the octopole, thereby closing an otherwise infinite system of moment equations in which the conductivity gradient couples the dynamics of neighboring moments.

2.2 Particle dynamics with frictional rolling

Neglecting the effects of particle inertia, the particle dynamics is determined by the balance of electric, hydrodynamic, and frictional forces and torques

$$\begin{aligned}\mathbf{F}_\mathrm{e} + \mathbf{F}_\mathrm{h} + \mathbf{F}_\mathrm{f} &= 0\\ \mathbf{T}_\mathrm{e} + \mathbf{T}_\mathrm{h} + \mathbf{T}_\mathrm{f} &= 0\end{aligned} \tag{2.23}$$

Here, the electric force $\mathbf{F}_\mathrm{e}$ and torque $\mathbf{T}_\mathrm{e}$ are related to the charge moments and the external potential gradients evaluated at the particle center

$$\begin{aligned}\mathbf{F}_\mathrm{e} &= -4\pi\varepsilon_\mathrm{f}\big(q\nabla\varphi_\mathrm{e}(\mathbf{0}) + \mathbf{p}\cdot\nabla\nabla\varphi_\mathrm{e}(\mathbf{0}) + \tfrac{1}{6}\mathbf{Q}[\cdot]^3\nabla\nabla\nabla\varphi_\mathrm{e}(\mathbf{0}) + \dots\big)\\ \mathbf{T}_\mathrm{e} &= -4\pi\varepsilon_\mathrm{f}\big(\mathbf{p}\times\nabla\varphi_\mathrm{e}(\mathbf{0}) + \tfrac{1}{3}(\mathbf{Q}\cdot\nabla)\times\nabla\varphi_\mathrm{e}(\mathbf{0}) + \dots\big)\end{aligned} \tag{2.24}$$

At low Reynolds number, the hydrodynamic force $\mathbf{F}_\mathrm{h}$ and torque $\mathbf{T}_\mathrm{h}$ on a particle in a quiescent fluid are linearly related to the it's linear and angular velocity as

$$\begin{bmatrix}\mathbf{U}\\ \mathbf{\Omega}\end{bmatrix} = -\begin{bmatrix}\mathbf{a} & \tilde{\mathbf{b}}\\ \mathbf{b} & \mathbf{c}\end{bmatrix}\cdot\begin{bmatrix}\mathbf{F}_\mathrm{h}\\ \mathbf{T}_\mathrm{h}\end{bmatrix} \tag{2.25}$$

For the sphere-wall geometry, the mobility tensors $\mathbf{a}$, $\mathbf{b}$, etc. have the axisymmetric forms[5]

$$\begin{aligned}a_{ij} &= (6\pi\eta a)^{-1}\big[Y_a(\delta_{ij}-\delta_{i3}\delta_{j3}) + X_a\delta_{i3}\delta_{j3}\big]\\ b_{ij} = -\tilde{b}_{ij} &= (6\pi\eta a^2)^{-1}\big[-Y_b\varepsilon_{ij3}\big]\\ c_{ij} &= (6\pi\eta a^3)^{-1}\big[Y_c(\delta_{ij}-\delta_{i3}\delta_{j3}) + X_c\delta_{i3}\delta_{j3}\big]\end{aligned} \tag{2.26}$$

where the mobility coefficients Y_a, X_a, etc. depend on the sphere-wall surface separation ℓ scaled by the particle radius a. We assume a constant surface separation of $\ell = 0.001a$ for which the relevant mobility coefficients are $Y_a = 0.22055$, $Y_b = 0.03508$, and $Y_c = 0.24459$.[6]

The frictional force $\mathbf{F}_\mathrm{f}$ and torque $\mathbf{T}_\mathrm{f}$ are chosen to maintain the following kinematic condition for a single particle in a quiescent fluid above a plane wall

$$\mathbf{U} = \lambda a\mathbf{\Omega}\times\mathbf{e}_3 \tag{2.27}$$

Here, λ is a dimensionless parameter that characterizes the strength of rotation-translation coupling: $\lambda = 1$ for no-slip frictional rolling; $\lambda = Y_b/Y_c \le 0.25$ for force-free "rolling" in the absence of frictional interactions. We adopt an intermediate value of $\lambda = 0.6$ to reproduce the rolling speeds and oscillation frequencies observed in experiment (see Figure S4 below).[7] The frictional force acts at the point of contact between the sphere and the wall, resulting in a frictional torque

$$\mathbf{T}_\mathrm{f} = -\mathbf{e}_3\times\mathbf{F}_\mathrm{f} \tag{2.28}$$

Substituting equations (2.28) and (2.27) into the mobility relation (2.25), the resulting friction force is

$$\begin{aligned}\mathbf{F}_\mathrm{f} = -&\left[\left(\frac{Y_a-\lambda Y_b}{Y_a+\lambda Y_c-(\lambda+1)Y_b}\right)(\boldsymbol{\delta}-\mathbf{e}_3\mathbf{e}_3)+\mathbf{e}_3\mathbf{e}_3\right]\cdot\mathbf{F}_\mathrm{e}\\ &-\left(\frac{\lambda Y_c-Y_b}{Y_a+\lambda Y_c-(\lambda+1)Y_b}\right)\frac{1}{a}\mathbf{e}_3\times\mathbf{T}_\mathrm{e}\end{aligned} \tag{2.29}$$

where the parenthetical terms involving the mobility coefficients and the slip parameter λ take the values 0.6411 and 0.3589, respectively, for $\lambda = 0.6$ and the chosen surface separation of $\ell/a = 0.001$. For collections of interacting particles, this friction equation is maintained, thereby perturbing the kinematic condition (2.27) due to additional hydrodynamic forces and torques.

For a single sphere in a quiescent fluid, the application of an electric torque in the 2-direction drives rotation and translation like

$$\Omega_2 = \frac{Y_c' T_2^\mathrm{e}}{6\pi\eta a^3} \quad\text{and}\quad U_1 = \lambda a\Omega_2 \tag{2.30}$$

where $Y_c' = (Y_aY_c - Y_b^2)/(Y_a+\lambda Y_c-(1+\lambda)Y_b)$ is a rotational mobility coefficient that accounts for frictional interactions with the substrate; $Y_c' = 0.1694$ for $\ell/a = 0.001$ and $\lambda = 0.6$.

2.3 Far field disturbance fields

Hydrodynamic disturbance. For a particle above a plane wall subject to a hydrodynamic force $\mathbf{F}_\mathrm{h}$ and torque $\mathbf{T}_\mathrm{h}$, the leading order hydrodynamic disturbance is that of a stokeslet, a rotlet, and their respective images[8,9]

$$\begin{aligned} u_i^\mathrm{d}(\mathbf{r}) = -\frac{F_j^\mathrm{h}}{8\pi\eta}\Bigg[&\left(\frac{\delta_{ij}}{r}+\frac{r_i r_j}{r^3}\right)-\left(\frac{\delta_{ij}}{R}+\frac{R_i R_j}{R^3}\right)+ \\ &2h(\delta_{j\alpha}\delta_{\alpha k}-\delta_{j3}\delta_{3k})\frac{\partial}{\partial R_k}\left(\frac{hR_i}{R^3}-\frac{\delta_{i3}}{R}-\frac{R_i R_3}{R^3}\right)\Bigg] \\ -\frac{T_j^\mathrm{h}}{8\pi\eta}&\left[\frac{\epsilon_{ijk}r_k}{r^3}+\frac{\epsilon_{ijk}R_k}{R^3}+2h\epsilon_{kj3}\left(\frac{\delta_{ik}}{R^3}-\frac{3R_iR_k}{R^5}\right)+6\epsilon_{kj3}\frac{R_iR_kR_3}{R^5}\right]+\dots \end{aligned} \tag{2.31}$$

where $h = a + \ell \approx a$ is the height of the particle above the wall, $\mathbf{R} = \mathbf{r} + 2h\mathbf{e}_3$ is the displacement from its image, and the index α takes only the values 1 or 2. Here, the first term is the stokeslet disturbance of strength $-\mathbf{F}_\mathrm{h}$ in an unbounded fluid; the second is that of an image stokeslet of strength $\mathbf{F}_\mathrm{h}$ at $\mathbf{r} = -2h\mathbf{e}_3$. The remaining contributions include a Stokes doublet and a source doublet such that the no slip condition is satisfied on the surface of the plane wall.[8] This expression for the disturbance velocity neglects higher order contributions due to the stresslet, etc. Figure S2 shows the stokeslet and rotlet flows for unit forces are torques in the plane of the surface.

The leading order contribution to the disturbance field in the plane of the particle is given by

$$\mathbf{u}_\mathrm{d}(\mathbf{r}) = -\frac{12h^2}{8\pi\eta}\frac{\mathbf{r}\mathbf{r}\cdot\mathbf{F}_\mathrm{h}}{r^5}-\frac{6h}{8\pi\eta}\frac{\mathbf{r}(\mathbf{e}_3\times\mathbf{r})\cdot\mathbf{T}_\mathrm{h}}{r^5}+O(r^{-4}) \quad \text{for} \quad r_3 = 0 \tag{2.32}$$

Similarly, the velocity gradient $\nabla\mathbf{v}$ in the r_1r_2-plane can be approximated as

$$\nabla\mathbf{u}_\mathrm{d}(\mathbf{r}) = -\frac{12h}{8\pi\eta}\frac{\mathbf{e}_3\,\mathbf{r}\,\mathbf{r}\cdot\mathbf{F}_\mathrm{h}}{r^5}-\frac{6}{8\pi\eta}\frac{\mathbf{e}_3\,\mathbf{r}\,(\mathbf{e}_3\times\mathbf{r})\cdot\mathbf{T}_\mathrm{h}}{r^5}+O(r^{-4}) \quad \text{for} \quad r_3 = 0 \tag{2.33}$$

The linearized velocity field centered about the point $\mathbf{r}$ is a simple shear field of the form

$$\mathbf{u}_\mathrm{d}(\mathbf{y}) \approx \mathbf{u}_\mathrm{d}(\mathbf{r}) + (\mathbf{y}-\mathbf{r})\cdot\nabla\mathbf{u}_\mathrm{d}(\mathbf{r}) = (\dot\gamma_1\mathbf{e}_1+\dot\gamma_2\mathbf{e}_2)(h+y_3) \tag{2.34}$$

where the vector shear rate $\dot{\boldsymbol{\gamma}}$ in the r_1r_2-plane of the particle is

$$\dot{\boldsymbol{\gamma}} = -\frac{12h}{8\pi\eta}\frac{\mathbf{r}\mathbf{r}\cdot\mathbf{F}_\mathrm{h}}{r^5}-\frac{6}{8\pi\eta}\frac{\mathbf{r}(\mathbf{e}_3\times\mathbf{r})\cdot\mathbf{T}_\mathrm{h}}{r^5}+O(r^{-4}) \tag{2.35}$$

At the solid wall ($y_3 = -h$), the disturbance velocity is zero in accordance with the no-slip boundary condition.

Electric disturbance. The disturbance potential $\varphi_\mathrm{d}(\mathbf{r})$ due to the charge moments above the electrode surface is given by

$$\begin{aligned} \varphi_\mathrm{d}(\mathbf{r}) = \frac{q}{r}+\frac{r_ip_i}{r^3}+\frac{1}{2}\frac{r_ir_jQ_{ij}}{r^5}+\dots-\frac{q}{R}-\frac{R_i(\delta_{ik}-2\delta_{i3}\delta_{k3})p_k}{R^3} \\ -\frac{1}{2}\frac{R_iR_j(\delta_{ik}-2\delta_{i3}\delta_{k3})(\delta_{jl}-2\delta_{j3}\delta_{l3})Q_{kl}}{R^5}+\dots \end{aligned} \tag{2.36}$$

The first three terms describe the contributions of the charge, dipole, and quadrupole in an unbounded medium; the remaining terms describe the image contributions that ensure zero potential on the surface $r_3 = -h$.

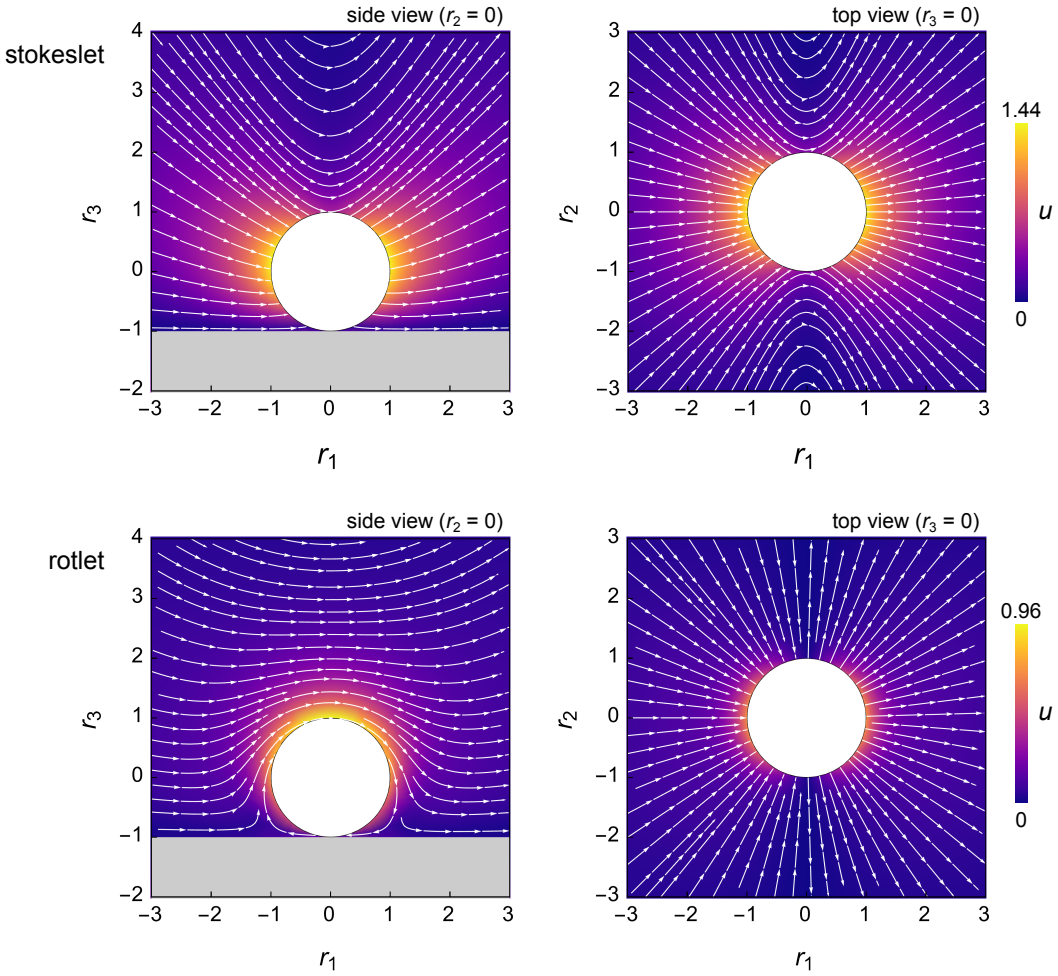

Figure S2: Fluid velocity in the 13-plane (left) and the 12-plane (right) for a unit stokeslet (top) in the 1-direction and a unit rotlet (bottom) in the 1-direction positioned a unit height above a plane wall at $r_3 = -1$.

In the far field, the leading order contributions to the electric potential in the plane of the particle is

$$\varphi_{\mathrm{d}}(\mathbf{r}) = \frac{2h(hq+p_3)}{r^3} + O(r^{-4}) \quad \text{for} \quad r_3 = 0 \tag{2.37}$$

Similarly, the disturbance field in the r_1r_2-plane is

$$-\nabla\varphi_{\mathrm{d}}(\mathbf{r}) = -\frac{2(hq+p_3)}{r^3}\mathbf{e}_3 + O(r^{-4}) \quad \text{for} \quad r_3 = 0 \tag{2.38}$$

The leading order contribution to the potential gradient gradient is

$$\nabla\nabla\varphi_{\mathrm{d}}(\mathbf{r}) = 0 + O(r^{-4}) \quad \text{for} \quad r_3 = 0 \tag{2.39}$$

2.4 Scales used for nondimensionalization

To facilitate our analysis, we adopt a convenient set of units in which the following quantities are set to one: the particle radius a, the fluid permittivity ε_{f}, the fluid conductivity $\kappa_{\mathrm{f}}(\mathbf{0})$, and the fluid viscosity η. These units correspond to the following characteristic scales used for nondimensionalization

$$\begin{aligned} \text{length} \quad & a \\ \text{time} \quad & \varepsilon_{\mathrm{f}}/\kappa_{\mathrm{f}}(\mathbf{0}) \\ \text{electric field} \quad & E_{\mathrm{s}} = \sqrt{\eta\kappa_{\mathrm{f}}(\mathbf{0})/\varepsilon_{\mathrm{f}}^2} \\ \text{charge} \quad & a^2E_{\mathrm{s}} \\ \text{dipole} \quad & a^3E_{\mathrm{s}} \\ \text{quadrupole} \quad & a^4E_{\mathrm{s}} \end{aligned} \tag{2.40}$$

We use the same notation to denote the resulting dimensionless variables. The dimensionless problem is fully specified by the external potential $\varphi_{\mathrm{e}}(\mathbf{r})$, the conductivity gradient $\nabla\kappa_{\mathrm{f}}(\mathbf{0})$, the particle permittivity ε_{p}, the particle conductivity κ_{p} (here zero), and the external flow field $\mathbf{u}_{\mathrm{e}}(\mathbf{r})$.

2.5 Single sphere in a uniform field

We first consider the dynamics of a single sphere in a uniform electric field, $\mathbf{E}_{\mathrm{e}} = E_{\mathrm{e}}\mathbf{e}_3$, and conductivity gradient, $\boldsymbol{\gamma} = \gamma\mathbf{e}_3$, both directed in the 3-direction (normal to the electrode). We assume that the particle rotates in the 2-direction such that the system has mirror symmetry about the 13-plane. The condition of mirror symmetry further requires that the components p_2,

Q_{12}, Q_{21}, Q_{23}, and Q_{32} are zero. With these simplifications, the model dynamics can be written as

$$\begin{aligned}
\dot{q} &= -q - \frac{\gamma}{3}(2p_3 + E_\mathrm{e}) \\
\dot{p_1} &= \Omega_2 \left(p_3 - \varepsilon'_\mathrm{cm} E_\mathrm{e}\right) - \frac{1}{\tau'} p_1 - \frac{3\gamma'}{5\tau'} Q_{13} \\
\dot{p_3} &= -\Omega_2 p_1 - \frac{1}{\tau'}(p_3 - \kappa'_\mathrm{cm} E_e) - \frac{\gamma'}{\tau'}\left(q + \tfrac{3}{5} Q_{33}\right) \\
\dot{Q}_{11} &= 2\Omega_2 Q_{13} - \frac{1}{\tau''} Q_{11} + \frac{2\gamma''}{3\tau''}(2p_3 + E_e) \\
\dot{Q}_{13} &= \Omega_2 (Q_{33} - Q_{11}) - \frac{1}{\tau''} Q_{13} - \frac{2\gamma''}{\tau''} p_1 \\
\dot{Q}_{33} &= -2\Omega_2 Q_{13} - \frac{1}{\tau''} Q_{33} - \frac{4\gamma''}{3\tau''}(2p_3 + E_e) \\
\dot{y_1} &= \lambda \Omega_2 = -\frac{2\lambda Y'_c}{3} p_1 E_\mathrm{e}
\end{aligned} \tag{2.41}$$

where y_1 is the particle position, and Y'_c is the effective mobility coefficient of equation (2.30) that accounts for friction. Other non-zero components of the symmetric and traceless quadrupole moment are related to those above as $Q_{31} = Q_{13}$ and $Q_{22} = -Q_{11} - Q_{33}$. The dimensionless parameters appearing in these equations depend on the particle permittivity ε_p and conductivity κ_p as

$$\begin{aligned}
\tau' &= \frac{\varepsilon_\mathrm{p} + 2}{\kappa_\mathrm{p} + 2}\,, \quad \gamma' = \frac{\gamma}{\kappa_\mathrm{p} + 2}\,, \quad \varepsilon'_\mathrm{cm} = \frac{\varepsilon_\mathrm{p} - 1}{\varepsilon_\mathrm{p} + 2}\,, \quad \kappa'_\mathrm{cm} = \frac{\kappa_\mathrm{p} - 1}{\kappa_\mathrm{p} + 2} \\
\tau'' &= \frac{2\varepsilon_\mathrm{p} + 3}{2\kappa_\mathrm{p} + 3}\,, \quad \gamma'' = \frac{\gamma}{2\kappa_\mathrm{p} + 3}\,, \quad \varepsilon''_\mathrm{cm} = \frac{\varepsilon_\mathrm{p} - 1}{2\varepsilon_\mathrm{p} + 3}\,, \quad \kappa''_\mathrm{cm} = \frac{\kappa_\mathrm{p} - 1}{2\kappa_\mathrm{p} + 3}
\end{aligned} \tag{2.42}$$

The problem is therefore specified by four dimensionless parameters: the particle permittivity ε_p, the particle conductivity κ_p, the external field E_e, and conductivity gradient γ. This model neglects contributions of the image charge, dipole, and quadrupole described by equation (2.36) to the external potential $\varphi_\mathrm{e}(r)$ in the moment dynamics (2.17)–(2.19) and in the electric force/torque (2.24). These contributions could be added; however, the resulting model would still be approximate. The truncated multipole expansion—valid in the far field—cannot accurately describe near field interactions with the electrode surface.[10] Moreover, the simplified model neglecting image interactions has been studied previously to identify the conditions necessary for Quincke oscillations.[1] Figure S3 shows the model dynamics of a single sphere in the oscillatory regime.[1] Figure S4 compares model predictions to previous measurements of the rolling speed and oscillation frequency for the relevant experimental conditions.[1]

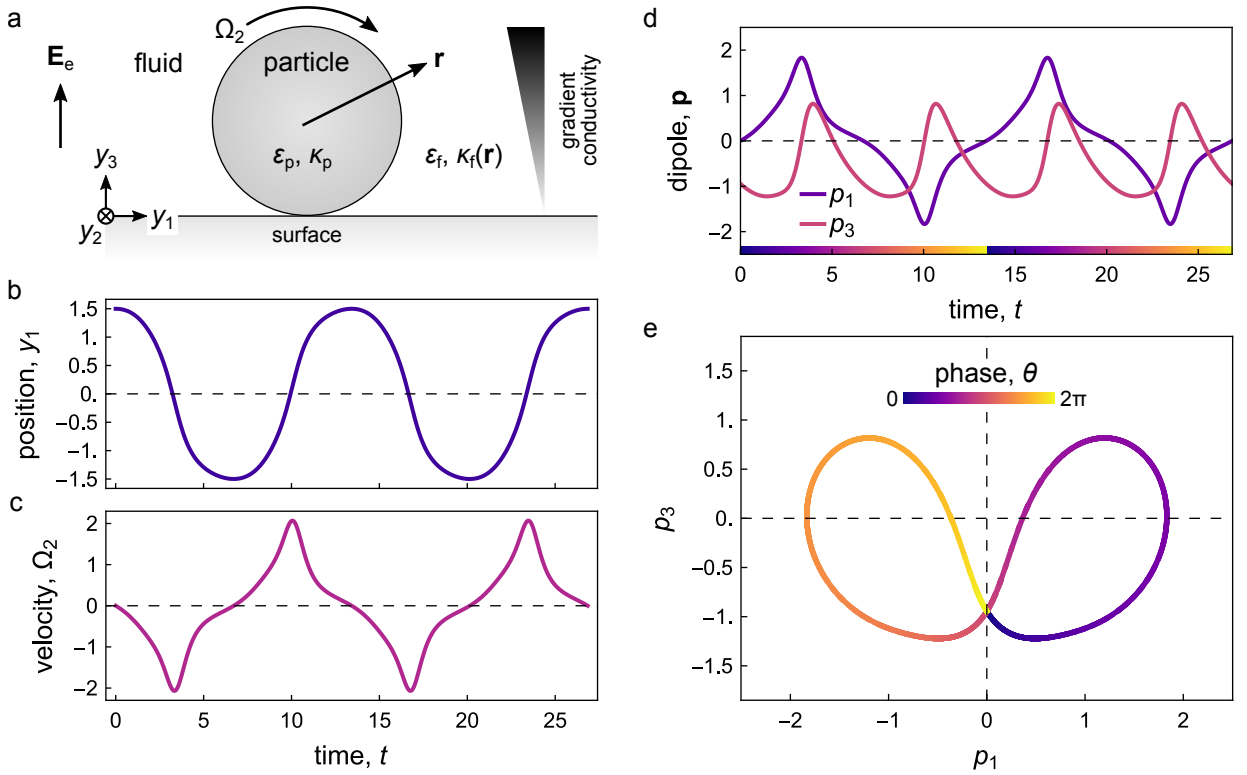

Figure S3: (a) Schematic illustration of a single spherical particle of radius a, permittivity ε_p, and conductivity κ_p immersed in a viscous fluid of permittivity ε_f and conductivity $\kappa_f(\mathbf{r})$. Different from the standard model of Quincke rotation, we consider the effects of conductivity gradients $\nabla\kappa_f \neq 0$, which describe asymmetric rates of charge accumulation at the particle surface.[7] (b-e) Model dynamics showing stable particle oscillations for (dimensionless) parameters $\varepsilon_p = 1.2$, $\kappa_p = 0$, $E_e = 10$, $\gamma = 1.22$. The angular oscillation frequency is $\omega = 0.468$. All quantities are presented using natural units described in Section 2.4.

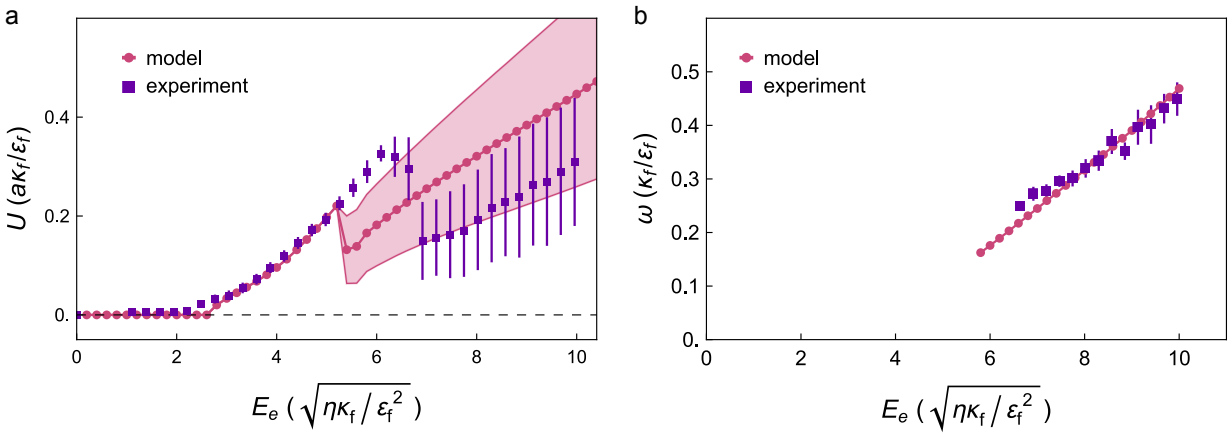

Figure S4: Comparing model predictions with experimental observations. (a) Time-averaged particle speed U vs. external field strength E_e as measured experimentally[7] (blue squares) and predicted by the dynamical model (pick circles). The shaded region and the error bars denote one standard deviation about the mean; large variations in speed indicate oscillatory dynamics. (b) Oscillation frequency ω vs. external field strength E_e. Model predictions (pink curves) are computed for $\varepsilon_\mathrm{p} = 1.17\varepsilon_\mathrm{f}$, $\kappa_\mathrm{p} = 0$, $\gamma = 1.22$, $\lambda = 0.6$. Quantities are presented using natural units detailed in the main text. Experimental data is for polystyrene spheres with radius $a = 5$ μm and permittivity $\varepsilon_\mathrm{p} = 2.4\varepsilon_0$ in a 0.15 M AOT-hexadecane solution.[7] With fluid viscosity $\eta = 2.9$ mPa·s, permittivity $\varepsilon_\mathrm{f} = 2.05\varepsilon_0$, and conductivity $\kappa_\mathrm{f} = 4.1 \times 10^{-8}$ S/m, the characteristic field strength and time scale used for nondimensionalization are $(\eta\kappa_\mathrm{f}/\varepsilon_\mathrm{f}^2)^{1/2} = 0.60$ V/μm and $\varepsilon_\mathrm{f}/\kappa_\mathrm{f} = 0.44$ ms. The conductivity gradient γ and rotation-translation coupling λ are tuned to reduce discrepancies between model and experiment.

Supplementary Note 3: SD simulations of two interacting oscillators

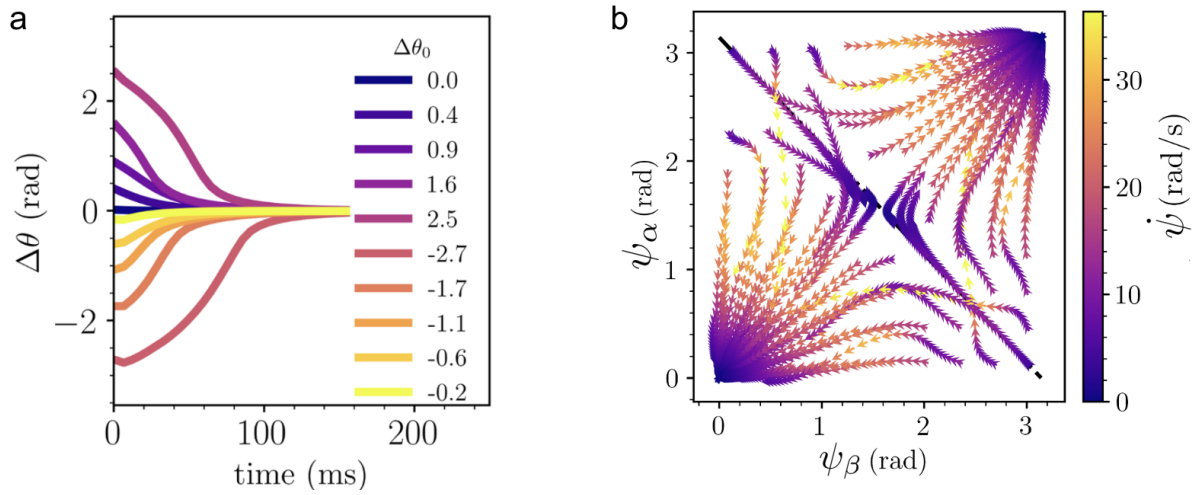

Figure S5: (a) Time evolution of the phase difference, $\Delta\theta$, between two oscillators separated by a distance $d = 3a$. Starting from an initial phase offset $\Delta\theta_0$, the particles rapidly synchronize within approximately 100 ms. (b) Phase portraits showing the evolution of the oscillation angles $(\psi_\alpha, \psi_\beta)$ for different initial orientations of synchronized but misaligned particles. The angles $\psi_\alpha = \chi_\alpha - \mu$ measure each particle's oscillation axis relative to the line joining their centers, where μ denotes the azimuthal angle of the interparticle axis. In all cases, the particles reach full alignment within 200 ms. The points $(\psi_\alpha, \psi_\beta) = (0, 0)$ and (π, π) correspond to equivalent steady states in which both particles oscillate along a common axis. The diagram shows that these states are stable. Eliminating dipolar interactions yields quantitatively similar results.

Supplementary Note 4: Fluid structure and ordering in SD simulations

The simulations reproduce the same structural and dynamical features observed in experiments. As shown in Fig. S6a, the oscillators form a fluid-like arrangement with short-range positional order. Their phases and orientations are correlated at $(\Delta\theta, \Delta\chi) = (0, 0)$, with additional peaks at (π, π), $(-\pi, \pi)$, $(-\pi, -\pi)$, and $(\pi, -\pi)$ due to phase-angle symmetry (Fig. S6b). As in the experiments, the oscillation frequency increases with particle density (Fig. S6c), though the effect is less pronounced in simulations. This difference is attributed to differences in the particle distributions: simulated particles are initialized with a uniform distribution, whereas the experiments exhibit local density fluctuations that enhance (reduce) the oscillation frequency in denser (sparser) regions.

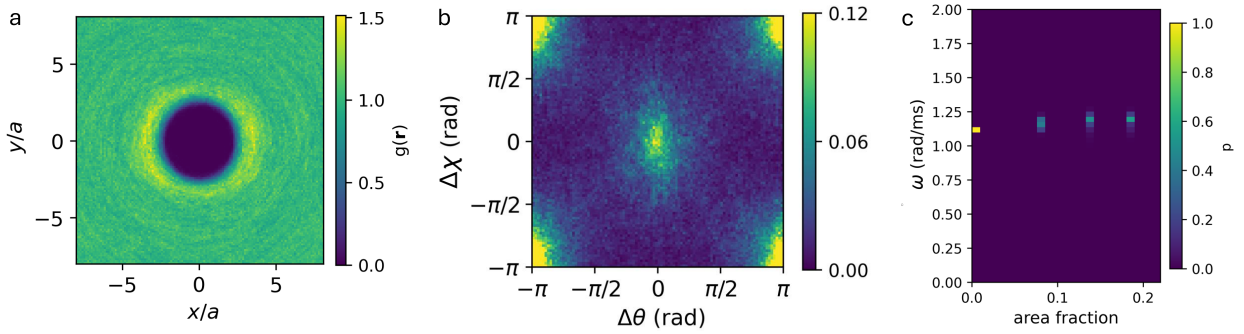

Figure S6: (a) Pair correlation function $g(\mathbf{r})$ as a function of displacements parallel x and perpendicular y to the oscillation direction. (b) Time-averaged histogram of the angle difference, $\Delta\chi = \chi_\beta - \chi_\alpha$, and the phase difference, $\Delta\theta = \theta_\beta - \theta_\alpha$ for oscillator pairs α and β separated by one to two particle diameters. (c) Oscillation frequency ω as a function of particle area fraction ρ.

Supplementary Note 5: Synchronematic correlation length in SD simulations

To characterize the phase–alignment order parameter in the simulations, the observed exponential decay is fitted to the function $S_{\chi\theta} = a\exp(-r/\ell) + S^{\infty}_{\chi\theta}$, where ℓ is the spatial correlation length, and $S^{\infty}_{\chi\theta}$ represents the asymptotic value of the correlation. If $S^{\infty}_{\chi\theta} \neq 0$, the system exhibits a certain degree of long-range ordering. In Fig. S7, we show how this spatial correlation length behaves for different system sizes and in the absence of dipolar interactions. For simulation sizes of $N = 200$ and 400 particles, five repetitions were performed to characterize the variability of the correlation length for different random initial conditions. For the larger system size, $N = 600$, only a single simulation was carried out at each point due to the high computational cost. The shaded areas for $N = 200$ and 400 represent the standard deviation of the measurements.

The correlation lengths are always close to the experimental range within the error bars, showing quantitative agreement for smaller diffusion coefficients. Figure S7a shows two main trends: an increase in the standard deviation for smaller diffusion coefficients and a decrease in correlation length with increasing diffusion coefficient. The first trend reflects the higher influence of the specific spatial configuration of the oscillators at lower diffusion coefficients. When the particles are randomly distributed in the simulation box, the centers of oscillation become fixed after a transient time. Thus, smaller diffusion coefficients make the system more likely to freeze into configurations that yield different correlation lengths. As noise increases, the standard deviation decreases, indicating that the initial configuration becomes less significant.

The asymptotic value of the exponential decay, shown in Fig. S7b, follows the same trends as in Fig. S7a and provides additional information about long-range correlations. For diffusion coefficients $D_\chi > 4.5$ rad^2/s, the system shows no long-range correlation.

Figure S7c compares simulations with and without dipolar interactions. Interestingly, the correlation lengths are quantitatively similar, indicating that the origin of this correlation length is primarily hydrodynamic. Furthermore, Fig. S7d shows that, in the absence of dipolar interactions, the system exhibits slightly shorter correlations.

When both dipolar and hydrodynamic interactions are removed, the simulation results become trivial: because the system is not dense enough for excluded-volume interactions, the oscillations occur independently and the phase–angle dynamics of different oscillators remain uncorrelated.

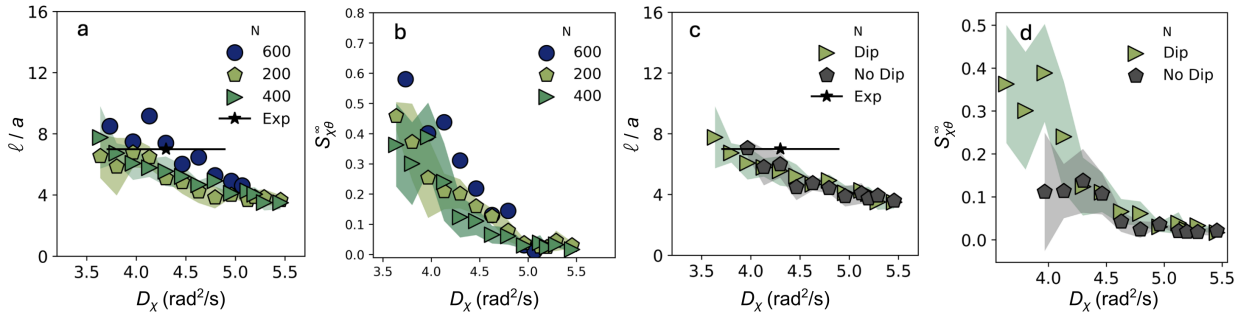

Figure S7: Characterization of correlation length in simulations. All simulations are performed at an area fraction $\rho = 0.14$. (a) Correlation length as a function of the rotational diffusion coefficient for numbers system sizes with N particles. (b) Stabilized bulk correlation length for different system sizes. (c) Comparison of simulations with and without dipolar interactions for a simulation with $N = 400$ particles. (d) Comparison bulk correlation length with and without dipolar interactions for a simulation with $N = 400$ particles.

Taken together, these results highlight hydrodynamic interactions as the main mechanism driving order in the fluid phase.

Supplementary Note 6: Model dynamics of two Quincke oscillators

6.1 Two spheres in a uniform field

We now consider the dynamics of two interacting spheres (denoted α and β) subject to a uniform field $\mathbf{E}_e = E_e\mathbf{e}_3$ and conductivity gradient $\nabla\kappa_\mathrm{f} = \gamma\mathbf{e}_3$ in the 3-direction (Fig. S8a). The spheres are located at positions $\mathbf{y}_\alpha$ and $\mathbf{y}_\beta$ in the plane $y_3^\alpha = y_3^\beta = h = 1.001$ normal to the field. Each sphere induces electric and hydrodynamic disturbance fields that influence the dynamics of the other.

Hydrodynamic interactions. Hydrodynamic interactions between two equally sized spheres above a plane wall are described using the Stokesian Dynamics (SD) method of Swan and Brady.[11] This approach starts with a far-field approximation of the mobility tensor $\mathcal{M}^\infty$ based on multipole expansions centered on each particle α and β

$$-\begin{bmatrix}\mathcal{M}^\infty_{\alpha\alpha} & \mathcal{M}^\infty_{\alpha\beta}\\ \mathcal{M}^\infty_{\beta\alpha} & \mathcal{M}^\infty_{\beta\beta}\end{bmatrix}\cdot\begin{bmatrix}\mathcal{F}^\alpha_\mathrm{h}\\ \mathcal{F}^\beta_\mathrm{h}\end{bmatrix} = \begin{bmatrix}\mathcal{U}^\alpha\\ \mathcal{U}^\beta\end{bmatrix} \tag{6.43}$$

Here, $\mathcal{F}^\alpha_\mathrm{h}$ denotes the moments of the force density on the surface of particle α (i.e., the force, torque, stresslet, etc.); similarly, $\mathcal{U}^\alpha$ denotes the velocity moments of particle α (namely, $\mathbf{U}^\alpha$ and $\mathbf{\Omega}^\alpha$ for rigid particles). We focus on the limit of widely separated particles and consider only the leading order terms of the pair mobility tensors $\mathcal{M}^\infty_{\alpha\beta} = (\mathcal{M}^\infty_{\beta\alpha})^\top$. As shown below, these tensors are of order r^{-3}, and we neglect higher order contributions.

Importantly, these far-field interactions between the widely separated spheres are modified by their respective interactions with the nearby wall. In the SD approach, these effects are included by adding sphere-wall corrections to the inverse of the far-field mobility tensor

$$\mathcal{R} = (\mathcal{M}^\infty)^{-1} + \mathcal{R}^\mathrm{w} - \mathcal{R}^{\mathrm{w},\infty} \tag{6.44}$$

where $\mathcal{R}^\mathrm{w}$ is the block diagonal sphere-wall resistance tensor describing near-field lubrication effects, and $\mathcal{R}^\mathrm{w}$ is that part of the sphere-wall interaction included in the far-field approximation (to avoid double counting).

In the limit of widely separated particles, the inverse of the far-field mobility tensor can be approximated as

$$(\mathcal{M}^\infty)^{-1} = \begin{bmatrix}(\mathcal{M}^\infty_{\alpha\alpha})^{-1} & -(\mathcal{M}^\infty_{\alpha\alpha})^{-1}\mathcal{M}^\infty_{\alpha\beta}(\mathcal{M}^\infty_{\beta\beta})^{-1}\\ -(\mathcal{M}^\infty_{\beta\beta})^{-1}\mathcal{M}^\infty_{\beta\alpha}(\mathcal{M}^\infty_{\alpha\alpha})^{-1} & (\mathcal{M}^\infty_{\beta\beta})^{-1}\end{bmatrix} + \mathcal{O}(r^{-4}) \tag{6.45}$$

The addition of the sphere-wall correction along the diagonal results in the exact sphere-wall resistance tensor for each particle

$$\mathcal{R} = \begin{bmatrix}\mathcal{R}_{\alpha\alpha} & -(\mathcal{M}^\infty_{\alpha\alpha})^{-1}\mathcal{M}^\infty_{\alpha\beta}(\mathcal{M}^\infty_{\beta\beta})^{-1}\\ -(\mathcal{M}^\infty_{\beta\beta})^{-1}\mathcal{M}^\infty_{\beta\alpha}(\mathcal{M}^\infty_{\alpha\alpha})^{-1} & \mathcal{R}_{\beta\beta}\end{bmatrix} + \mathcal{O}(r^{-4}) \tag{6.46}$$

Inverting this result, the wall-corrected mobility tensor is estimated as

$$\mathcal{M} = \begin{bmatrix}\mathcal{M}_{\alpha\alpha} & \mathcal{M}_{\alpha\alpha}(\mathcal{M}^\infty_{\alpha\alpha})^{-1}\mathcal{M}^\infty_{\alpha\beta}(\mathcal{M}^\infty_{\beta\beta})^{-1}\mathcal{M}_{\beta\beta}\\ \mathcal{M}_{\beta\beta}(\mathcal{M}^\infty_{\beta\beta})^{-1}\mathcal{M}^\infty_{\beta\alpha}(\mathcal{M}^\infty_{\alpha\alpha})^{-1}\mathcal{M}_{\alpha\alpha} & \mathcal{M}_{\beta\beta}\end{bmatrix} + \mathcal{O}(r^{-4}) \tag{6.47}$$

where $\mathcal{M}_{\alpha\alpha} = \mathcal{R}_{\alpha\alpha}$ is the exact sphere-wall mobility tensor a single sphere. The evaluation of the leading order hydrodynamic interactions captured by the pair mobility tensors $\mathcal{M}_{\alpha\beta} = (\mathcal{M}_{\beta\alpha})^\top$

requires the far-field pair tensors $\mathcal{M}^{\infty}_{\alpha\beta} = (\mathcal{M}^{\infty}_{\beta\alpha})^{\top}$, far-field "self" tensors $\mathcal{M}^{\infty}_{\alpha\alpha} = \mathcal{M}^{\infty}_{\beta\beta}$, and the exact single sphere-wall tensors $\mathcal{M}_{\alpha\alpha} = \mathcal{M}_{\beta\beta}$.

For a single sphere α in a quiescent fluid above a plane wall, the exact mobility tensor relates the hydrodynamic force, torque, and stresslet to the linear and angular velocity of the particle as

$$\begin{bmatrix} \mathbf{U} \\ \mathbf{\Omega} \\ \mathbf{0} \end{bmatrix} = -\mathcal{M}_{\alpha\alpha} \cdot \begin{bmatrix} \mathbf{F}_{\mathrm{h}} \\ \mathbf{T}_{\mathrm{h}} \\ \mathbf{S}_{\mathrm{h}} \end{bmatrix} \tag{6.48}$$

For simplicity, we neglect the stresslet contributions in the present analysis and in the simulations reported in the main text. With this simplification, the mobility relation is given by equations (2.25) and (2.26) of the previous section. In addition to the exact sphere-wall mobility coefficients Y_a, Y_b, and Y_c,[6] we evaluate the approximate far-field mobility coefficients $Y_a^{\infty} = 0.500499$, $Y_b^{\infty} = 0.0933759$, and $Y_c^{\infty} = 0.516327$ for $h = 1.001a$ using the "self" mobility expressions of Swan and Brady.[11]

For two widely separated spheres α and β in a common plane $y_3^{\alpha} = y_3^{\beta} = h$ above a wall, the pair mobility tensors have the following asymptotic forms for large r

$$\begin{aligned}
\mathbf{a}^{\infty}_{\alpha\beta} = (\mathbf{a}^{\infty}_{\beta\alpha})^{\top} &= \frac{12h^2}{8\pi r^3} \begin{bmatrix} \hat{r}_1^2 & \hat{r}_1\hat{r}_2 & 0 \\ \hat{r}_1\hat{r}_2 & \hat{r}_2^2 & 0 \\ 0 & 0 & 0 \end{bmatrix} + O(r^{-4}) \\
\mathbf{b}^{\infty}_{\alpha\beta} = (\tilde{\mathbf{b}}^{\infty}_{\beta\alpha})^{\top} = (\tilde{\mathbf{b}}^{\infty}_{\alpha\beta})^{\top} = \mathbf{b}^{\infty}_{\beta\alpha} &= \frac{6h}{8\pi r^3} \begin{bmatrix} -\hat{r}_1\hat{r}_2 & -\hat{r}_2^2 & 0 \\ \hat{r}_1^2 & \hat{r}_1\hat{r}_2 & 0 \\ 0 & 0 & 0 \end{bmatrix} + O(r^{-4}) \\
\mathbf{c}^{\infty}_{\alpha\beta} = (\mathbf{c}^{\infty}_{\beta\alpha})^{\top} &= \frac{3}{8\pi r^3} \begin{bmatrix} \hat{r}_2^2 & -\hat{r}_1\hat{r}_2 & 0 \\ -\hat{r}_1\hat{r}_2 & \hat{r}_1^2 & 0 \\ 0 & 0 & 0 \end{bmatrix} + O(r^{-4})
\end{aligned} \tag{6.49}$$

where $\mathbf{r} = \mathbf{y}_{\beta} - \mathbf{y}_{\alpha}$ is the displacement from sphere α to β with magnitude $r = \sqrt{\mathbf{r}\cdot\mathbf{r}}$ and direction $\hat{\mathbf{r}} = \mathbf{r}/r$. The wall-corrected mobility tensor $\mathcal{M}_{\alpha\beta} = \mathcal{M}_{\alpha\alpha}(\mathcal{M}^{\infty}_{\alpha\alpha})^{-1}\mathcal{M}^{\infty}_{\alpha\beta}(\mathcal{M}^{\infty}_{\beta\beta})^{-1}\mathcal{M}_{\beta\beta}$ has components that share the functional forms of the far field approximations but with renormalized magnitudes

$$\begin{aligned}
\mathbf{a}_{\alpha\beta} = (\mathbf{a}_{\beta\alpha})^{\top} &= f_a^2\, \mathbf{a}^{\infty}_{\alpha\beta} \quad \text{with} \quad f_a = \frac{Y_b Y_a^{\infty} - (Y_a + 2hY_b)Y_b^{\infty} + 2hY_aY_c^{\infty}}{2(Y_a^{\infty}Y_c^{\infty} - Y_b^{\infty 2})} \\
\mathbf{b}_{\alpha\beta} = (\tilde{\mathbf{b}}_{\beta\alpha})^{\top} = (\tilde{\mathbf{b}}_{\alpha\beta})^{\top} &= \mathbf{b}_{\beta\alpha} = f_a f_c\, \mathbf{b}^{\infty}_{\alpha\beta} \\
\mathbf{c}_{\alpha\beta} = (\mathbf{c}_{\beta\alpha})^{\top} &= f_c^2\, \mathbf{c}^{\infty}_{\alpha\beta} \quad \text{with} \quad f_c = \frac{Y_c Y_a^{\infty} - (Y_b + 2hY_c)Y_b^{\infty} + 2hY_bY_c^{\infty}}{Y_a^{\infty}Y_c^{\infty} - Y_b^{\infty 2}}
\end{aligned} \tag{6.50}$$

For the chosen surface separation $\ell = 1.001a$, the scaling factors are $f_a = 0.437292$ and $f_c = 0.439243$ such that the pair mobility tensors are all ca. 1/5 times the far field estimates.

Electric interactions. Electric interactions between the particles are computed in similar fashion to the hydrodynamic interactions with one distinction: the "self" sphere-wall interactions are not considered (see Section 2.5). Consequently, we include only the far-field "pair" interactions without the near-field wall corrections required for hydrodynamics. The electric field at the center of sphere α due to the external field and the disturbance field of sphere β is given by

$$-\nabla\varphi_{\mathrm{e}}(\mathbf{y}_{\alpha}) = \mathbf{E}_{\mathrm{e}} - \frac{2(hq^{\beta} + p_3^{\beta})}{r^3}\mathbf{e}_3 + O(r^{-4}) \tag{6.51}$$

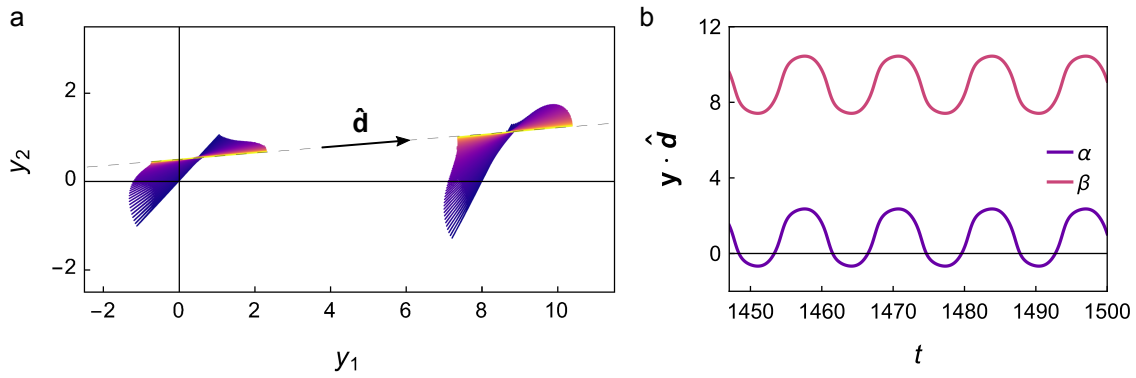

Figure S8: Numerical simulation results for two Quincke oscillators α and β for model parameters $\varepsilon_{\mathrm{p}} = 1.2$, $\kappa_{\mathrm{p}} = 0$, $E_{\mathrm{e}} = 10$, and $\gamma = 1.22$. Oscillator α is initialized at $\mathbf{x}_\alpha = [0, 0]^\top$ with phase $\theta_\alpha = 0$ and angle $\chi_\alpha = \pi/4$ measured relative to the 1-direction; oscillator β is initialized at $\mathbf{x}_\beta = [0, 8]^\top$ with phase $\theta_\beta = \pi/5$ and angle $\chi_\beta = \pi/3$. (a) The instantaneous positions $\mathbf{y}_\alpha$ and $\mathbf{y}_\beta$ of the spheres in the 12-plane evolve in time to align their back-and-forth oscillations along a common direction denoted by the unit vector $\hat{\mathbf{d}}$. (b) Ultimately, the particles synchronize their oscillations along the $\hat{\mathbf{d}}$ direction.

where $\mathbf{r} = \mathbf{y}_\beta - \mathbf{y}_\alpha$ is the displacement from sphere α to β. The potential gradient gradient is zero at this order. We substitute these expressions for the field and the potential gradient gradient into equations (2.17)–(2.19) for the moment dynamics and equation (2.24) for the electric force/torque, neglecting terms of order r^{-4} and higher. The dynamics of sphere β is similarly perturbed by the disturbance field due to sphere α.

Numerical Results. With these electrostatic and hydrodynamic interactions, the coupled dynamics of the two spheres can be integrated numerically to identify their stable behavior(s). For spheres separated by a large distance ($d = 10$) in the oscillating regime, we find one stable attractor in which particles align and synchronize their back-and-forth motions along a common direction (Figs. S8a,b). The distance between the two oscillators approaches a constant value since repulsive electrostatic interactions, which scale as r^{-4}, are not included in our treatment of leading order effects (Figs. S8c).

6.2 Weakly coupled particle oscillators

When the particles are sufficiently far apart, their coupled oscillatory dynamics can be accurately described using the formalism of weakly coupled oscillators.[12,13] In this approach, the dynamical state variable $Y(t)$ for each particle oscillator is replaced by a low-order approximation based on the oscillation phase $\theta(t)$, angle $\chi(t)$, and position $\mathbf{x}(t)$ (Fig. S9). Here, $Y(t)$ is an 11-dimensional vector that describes the charge $q(t)$, the three components of the dipole $\mathbf{p}(t)$, the five independent components of the quadrupole $\mathbf{Q}(t)$, and the two in-plane components of the position $\mathbf{y}(t)$. The dynamics of sphere α is expressed as

$$\dot{Y}_\alpha = F(Y_\alpha) + d^{-3} I_3(Y_\alpha, Y_\beta) + \mathcal{O}(d^{-4}) \tag{6.52}$$

where $F(Y_\alpha)$ describes the dynamics of the individual sphere, and the function $I_3(Y_\alpha, Y_\beta)$ describes leading order interactions with a neighboring sphere β. The distance d is the magnitude of the

displacement vector $\mathbf{d} = \mathbf{x}_\beta - \mathbf{x}_\alpha$ from oscillator α to β. The inverse distance d^{-1} plays the role of the small parameter in a singular perturbation analysis.

The oscillator phase θ describes its location in phase space along the limit cycle for a single particle

$$\theta(t) = (\omega t + \phi) \mod 2\pi \tag{6.53}$$

where $\omega = 2\pi/T$ is the oscillation frequency, and ϕ is the relative phase. We define the relative phase such that $\theta = 0$ corresponds to zero angular velocity prior to rotation in the negative 2-direction (see Fig. S3). The angle χ and position $\mathbf{x}$ describe the orientation and location of particle oscillations on the electrode surface. We define the angle $\chi = 0$ to describe oscillatory rotations about the 2-axis; the position $\mathbf{x} = 0$ describes back-and-forth motion about the origin (Fig. S9). Other limit cycles are obtained by rotation in the 3-direction by the angle χ and translation in the 12-plane to the position $\mathbf{x}$. For two oscillators separated by a displacement $\mathbf{d} = d\cos\mu\mathbf{e}_1 + d\sin\mu\mathbf{e}_2$, it will be convenient to parameterize the angles as $\psi_\alpha = \chi_\alpha - \mu$ and $\psi_\beta = \chi_\beta - \mu$, which are measured from the $\mathbf{d}$-direction rather than the $\mathbf{e}_1$-direction (Fig. S9).

To derive the leading order interaction function $I_3(Y_\alpha, Y_\beta)$ for two weakly coupled oscillators, we first decompose their transient positions as $\mathbf{y}(t) = \mathbf{x}(t) + \mathbf{z}(t)$, where $\mathbf{z}(t)$ describes rapid oscillatory displacements about the slowly varying position $\mathbf{x}(t)$ of the oscillation center (Fig. S9). Similarly, the instantaneous displacement from sphere α to β can be decomposed as $\mathbf{r} = \mathbf{d} + \boldsymbol{\epsilon}$, where $\boldsymbol{\epsilon} = \mathbf{z}_\beta - \mathbf{z}_\alpha$ describes rapid oscillations about the mean displacement $\mathbf{d}$. We expand the two-sphere dynamics of the previous section in a Taylor series in d retaining only the leading order $\mathcal{O}(d^{-3})$ contributions to determine the interaction function $I_3(Y_\alpha, Y_\beta)$.

Away from the limit cycle, we map each point $Y(t)$ in the basin of the limit cycle to the asymptotic phase to which it evolves: $\Phi(Y(t)) = \theta(t)$ as $t \to \infty$. Differentiating with respect to

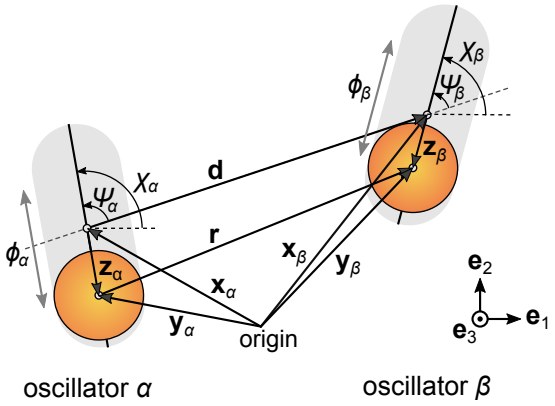

Figure S9: Schematic illustration of two particle oscillators α and β with instantaneous positions $\mathbf{y}_\alpha$ and $\mathbf{y}_\beta$. The particles (golden) oscillate back and forth within the respective grey regions. In the limit of weak coupling, particle motion is decomposed as $\mathbf{y}(t) = \mathbf{x}(t) + \mathbf{z}(t)$ where $\mathbf{z}(t)$ describes rapid oscillations about the slowly varying oscillation center $\mathbf{x}(t)$. The displacement vector between the particles is $\mathbf{r} = \mathbf{y}_\beta - \mathbf{y}_\alpha$; that between the oscillator centers is $\mathbf{d} = \mathbf{x}_\beta - \mathbf{x}_\alpha$. The orientation of the oscillatory motion $\mathbf{z}(t)$ is characterized by the angle χ measured clockwise relative to the 1-direction or the angle ψ measured relative to the $\mathbf{d}$-direction. The relative phase ϕ of each oscillator describes its progress through the periodic limit cycle.

time, we obtain the following equation for the evolution of the oscillator phase

$$\begin{aligned}\dot{\theta}_\alpha &= \nabla_Y \Phi(Y_\alpha) \cdot \dot{Y}_\alpha \\ &= \omega + d^{-3} \nabla_Y \Phi(Y_\alpha) \cdot I_3(Y_\alpha, Y_\beta) + \mathcal{O}(d^{-4})\end{aligned} \tag{6.54}$$

where we have made use of equation (6.52) and the fact that $\nabla_Y \Phi(Y_\alpha) \cdot F(Y_\alpha) = \omega$. For weakly coupled oscillators, the state variable $Y_\alpha(t)$ is well approximated by its value on the limit cycle $Y_\alpha(t) \approx Y_{\mathrm{LC}}(\omega t + \phi_\alpha, \psi_\alpha, \mathbf{x}_\alpha)$ as parameterized by the relative phase ϕ_α, the angle ψ_α, and the position $\mathbf{x}_\alpha$. Substituting this approximation into equation (6.54) and averaging the dynamics over one oscillation cycle, we obtain the following approximate dynamics for the relative phase of oscillator α

$$\begin{aligned}\dot{\phi}_\alpha &= \frac{1}{d^3 T} \int_0^T \nabla_Y \Phi\big(Y_{\mathrm{LC}}^\alpha(t)\big) \cdot I_3\big(Y_{\mathrm{LC}}^\alpha(t), Y_{\mathrm{LC}}^\beta(t)\big) dt + \cdots \\ &= d^{-3} H_3^\phi(\psi_\alpha, \psi_\beta, \Delta\phi) + \ldots\end{aligned} \tag{6.55}$$

where $Y_{\mathrm{LC}}^\alpha(t) \equiv Y_{\mathrm{LC}}(\omega t + \phi_\alpha, \psi_\alpha, \mathbf{x}_\alpha)$ denotes limit the cycle of oscillator α and likewise for $Y_{\mathrm{LC}}^\beta(t)$. The oscillation-averaged interaction function $H_3^\phi(\psi_\alpha, \psi_\beta, \Delta\phi)$ depends on the oscillator angles ψ_α and ψ_β and the phase difference $\Delta\phi = \phi_\beta - \phi_\alpha$.

We use the adjoint method[13,14] to compute the phase response curve (PRC) $\nabla_Y \Phi(Y_{\mathrm{LC}})$, which describes the shift in phase induced by an infinitesimal perturbation of the limit cycle. Briefly, the adjoint problem for $Z(t)$ is

$$\frac{dZ}{dt} = -J^T(t) Z \tag{6.56}$$

where $J(t) = \nabla_Y F\big(Y_{\mathrm{LC}}(\omega t, 0, \mathbf{0})\big)$ is the Jacobian matrix of partial derivatives evaluated on the canonical limit cycle with $\phi = \chi = 0$, and $\mathbf{x} = \mathbf{0}$. In this problem, there are four linearly independent solutions $Z_1(t), \ldots, Z_4(t)$ corresponding to Floquet modes of the periodic matrix $-J^T(t)$ with Floquet exponents of zero. These modes correspond to the four neutral degrees of freedom of the oscillator—namely, its phase, angle, and in-plane position. As derived elsewhere,[13,15] the slow dynamics of these quantities satisfy integral conditions of the form

$$\begin{aligned}0 = \int_0^T Z_i(t) \cdot \Big[& d^{-3} I_3\big(Y_{\mathrm{LC}}(\omega t, \chi_\alpha, \mathbf{x}_\alpha), Y_{\mathrm{LC}}(\omega t + \phi_\beta - \phi_\alpha, \chi_\beta, \mathbf{x}_\beta)\big) + \ldots \\ & - \partial_\phi Y_{\mathrm{LC}}(\omega t, \chi_\alpha, \mathbf{x}_\alpha) \dot{\phi}_\alpha - \partial_\chi Y_{\mathrm{LC}}(\omega t, \chi_\alpha, \mathbf{x}_\alpha) \dot{\chi}_\alpha - \nabla_{\mathbf{x}} Y_{\mathrm{LC}}(\omega t, \chi_\alpha, \mathbf{x}_\alpha) \cdot \dot{\mathbf{x}}_\alpha \Big] dt\end{aligned} \tag{6.57}$$

for $i = 1, \ldots, 4$. The time derivatives of the slow variables ϕ_α, χ_α, and $\mathbf{x}_\alpha$ are approximated as constant on the time scale of the oscillation period. Together, the four integral conditions (6.57) provide as many differential equations for the slow variables. Accounting for translational and rotational symmetries, the slow dynamics can be expressed as

$$\begin{aligned}\dot{\phi}_\alpha &= d^{-3}\, H_3^\phi(\psi_\alpha, \psi_\beta, \Delta\phi) + \mathcal{O}(d^{-4}) \\ \dot{\chi}_\alpha &= d^{-3}\, H_3^\chi(\psi_\alpha, \psi_\beta, \Delta\phi) + \mathcal{O}(d^{-4}) \\ \dot{\mathbf{x}}_\alpha &= 0 + \mathcal{O}(d^{-4})\end{aligned} \tag{6.58}$$

where the interaction functions are parameterized by the angles $\psi_\alpha = \chi_\alpha - \mu$ and $\psi_\alpha = \chi_\beta - \mu$ measured from the $\mathbf{d}$-direction and the phase difference $\Delta\phi = \phi_\beta - \phi_\alpha$. We include only the leading

order interactions, which are of order d^{-3} for the oscillator phase and angle. Similarly, the dynamics of oscillator β is governed by

$$\begin{aligned}\dot{\phi}_\beta &= d^{-3}\, H_3^\phi(\psi_\beta, \psi_\alpha, -\Delta\phi) + \mathcal{O}(d^{-4}) \\ \dot{\chi}_\beta &= d^{-3}\, H_3^\chi(\psi_\beta, \psi_\alpha, -\Delta\phi) + \mathcal{O}(d^{-4}) \\ \dot{\mathbf{x}}_\beta &= 0 + \mathcal{O}(d^{-4})\end{aligned} \tag{6.59}$$

6.3 Symmetries of the interaction functions

The interaction functions—for example, $H_3^\phi(\psi_\alpha, \psi_\beta, \Delta\phi)$—are 2π-periodic functions of the angles ψ_α, ψ_β and the phase difference $\Delta\phi = \phi_\beta - \phi_\alpha$

$$H(\psi_\alpha + 2\pi j, \psi_\beta + 2\pi k, \Delta\phi + 2\pi l) = H(\psi_\alpha, \psi_\beta, \Delta\phi) \tag{6.60}$$

for integers j, k, and l. Additionally, phase-angle symmetry of the limit cycle implies that a phase shift by π is equivalent to a change in angle by π

$$H(\psi_\alpha + \pi j, \psi_\beta + \pi k, \Delta\phi + (k-j)\pi) = H(\psi_\alpha, \psi_\beta, \Delta\phi) \tag{6.61}$$

for integers j and k. Finally, mirror symmetry of limit cycle implies the following relations upon reflection about a vertical plane containing the oscillator centers

$$\begin{aligned}H_3^\phi(-\psi_\alpha, -\psi_\beta, \Delta\phi) &= H_3^\phi(\psi_\alpha, \psi_\beta, \Delta\phi) \\ -H_3^\chi(-\psi_\alpha, -\psi_\beta, \Delta\phi) &= H_3^\chi(\psi_\alpha, \psi_\beta, \Delta\phi)\end{aligned} \tag{6.62}$$

Note that interaction functions for the phase are even with respect to this transformation; those for the angle are odd.

The above symmetries can be used to derive low-order approximations to the interaction functions. Expanding the interaction functions in a Fourier series and retaining only those terms that satisfy the required symmetries, we obtain the following approximation for the even interaction function

$$\begin{aligned}H_3^\phi(\psi_\alpha, \psi_\beta, \Delta\phi) = {} & A + B_1 \cos 2\psi_\alpha + B_2 \cos 2\psi_\beta + B_3 \sin 2\Delta\phi + B_4 \cos 2\Delta\phi \\ & + (C_1 \sin\Delta\phi + C_2 \cos\Delta\phi)\cos(\psi_\beta - \psi_\alpha) \\ & + (C_3 \sin\Delta\phi + C_4 \cos\Delta\phi)\cos(\psi_\beta + \psi_\alpha) \\ & + D_1 \cos(2\psi_\beta - 2\psi_\alpha) + D_2 \cos(2\psi_\beta + 2\psi_\alpha) \\ & + (E_1 \sin 3\Delta\phi + E_2 \cos 3\Delta\phi)\cos(\psi_\beta - \psi_\alpha) \\ & + (E_3 \sin 3\Delta\phi + E_4 \cos 3\Delta\phi)\cos(\psi_\beta + \psi_\alpha) + \ldots\end{aligned} \tag{6.63}$$

where the $A, B, C, ...$ are coefficients to be determined. Similarly, the odd interaction function can be approximated as

$$\begin{aligned}H_3^\chi(\psi_\alpha, \psi_\beta, \Delta\phi) = {} & B_1 \sin 2\psi_\alpha + B_2 \sin 2\psi_\beta \\ & + (C_1 \sin\Delta\phi + C_2 \cos\Delta\phi)\sin(\psi_\beta - \psi_\alpha) \\ & + (C_3 \sin\Delta\phi + C_4 \cos\Delta\phi)\sin(\psi_\beta + \psi_\alpha) \\ & + D_1 \sin(2\psi_\beta - 2\psi_\alpha) + D_2 \sin(2\psi_\beta + 2\psi_\alpha) \\ & + (E_1 \sin 3\Delta\phi + E_2 \cos 3\Delta\phi)\sin(\psi_\beta - \psi_\alpha) \\ & + (E_3 \sin 3\Delta\phi + E_4 \cos 3\Delta\phi)\sin(\psi_\beta + \psi_\alpha) + \ldots\end{aligned} \tag{6.64}$$

To estimate the coefficients, we first compute the interaction functions on a regular grid of 34×34×34 points in $\psi_\alpha, \psi_\beta, \Delta\phi$ space. The coefficients are estimated from these data using linear regression. This parametric approximation agrees with the numerically computed interaction function to within 1%.

6.4 Dynamics of two weakly coupled oscillators

Synchronization of in-line oscillators. For two in-line oscillators with angles $\psi_\alpha = \psi_\beta = 0$, the phase difference $\Delta\phi$ is governed by

$$\dot{\Delta\phi} = d^{-3}\big(H_3^\phi(0,0,-\Delta\phi) - H_3^\phi(0,0,\Delta\phi)\big) = d^{-3}G_3^\phi(0,0,\Delta\phi) \tag{6.65}$$

where d is constant at order d^{-3}. Figure S10a shows the $G_3^\phi(0,0,\Delta\phi)$ function and the relative contributions of electrostatic and hydrodynamic interactions. Phase coupling between neighboring oscillators is dominated by hydrodynamic interactions. The phase dynamics has one stable fixed point at $\Delta\phi = 0$ corresponding to in-phase synchronization. Similarly, the phase sum $\Sigma\phi = \phi_\alpha + \phi_\beta$ evolves as

$$\dot{\Sigma\phi} = d^{-3}\big(H_3^\phi(0,0,\Delta\phi) + H_3^\phi(0,0,-\Delta\phi)\big) = d^{-3}S_3^\phi(0,0,\Delta\phi) \tag{6.66}$$

where $S_3^\phi(0,0,\Delta\phi)$ describes changes in the common frequency of the two oscillators. Figure S10a shows that synchronized particles oscillate faster than individual particles by an additive correction of $5.95d^{-3}$. These predictions for weakly coupled oscillators agree quantitatively with full numerical solutions for two widely separated particles $d \gg 1$ (Fig. S10b).

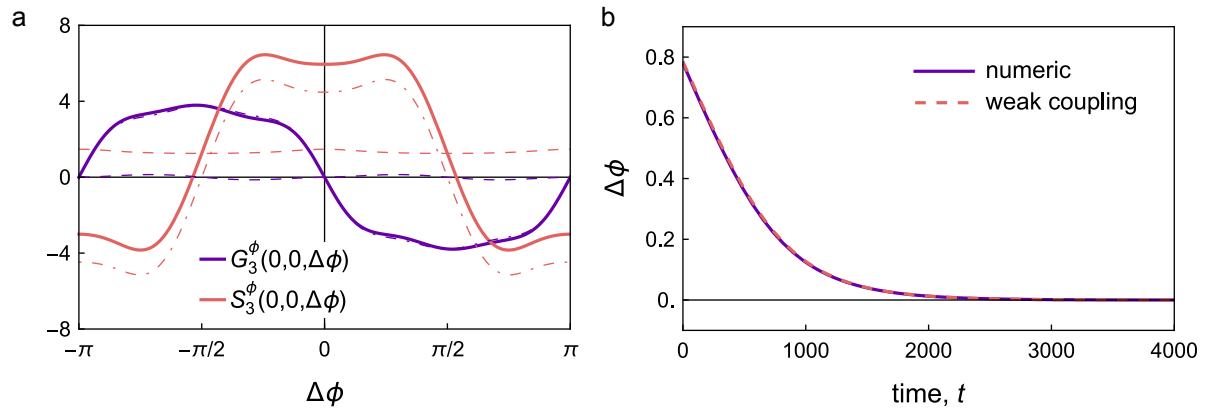

Figure S10: (a) Interaction functions $G_3^\phi(0,0,\Delta\phi)$ and $S_3^\phi(0,0,\Delta\phi)$ governing the evolution of the phase difference $\Delta\phi$ and phase sum, respectively. Dashed and dash-dotted curves denote electrostatic and hydrodynamic contributions to the total interactions (solid curves). (b) The phase difference $\Delta\phi$ between two in-line oscillators with $\psi_\alpha = \psi_\beta = 0$ relaxes to zero at a characteristic rate of $7.53d^{-3} + \mathcal{O}(d^{-4})$. The dashed curve shows the prediction of equation (6.65) for two weakly coupled oscillators separated by a distance $d = 10$ with an initial phase difference $\Delta\phi(0) = \pi/4$; the solid curve shows the full numerical solution.

Alignment of in-phase oscillators. For two in-phase oscillators with $\Delta\phi = 0$, the angles ψ_α and ψ_β evolve as

$$\begin{aligned} \dot{\psi_\alpha} &= d^{-3} H_3^\chi(\psi_\alpha, \psi_\beta, 0) \\ \dot{\psi_\beta} &= d^{-3} H_3^\chi(\psi_\beta, \psi_\alpha, 0) \end{aligned} \tag{6.67}$$

Figure S11a shows the phase portrait of the angle dynamics, which has a stable fixed point at the origin. In-phase oscillators align their back-and-forth motions along the line of centers at a characteristic rate $2.68d^{-3}$. At this order, the angle interaction function $H_3^\chi(\psi_\alpha, \psi_\beta, \Delta\phi)$ is due entirely to hydrodynamic interactions between neighboring oscillators. These predictions of equations (6.67) for weakly coupled oscillators agree quantitatively with full numerical solutions for two widely separated particles (Fig. S11b).

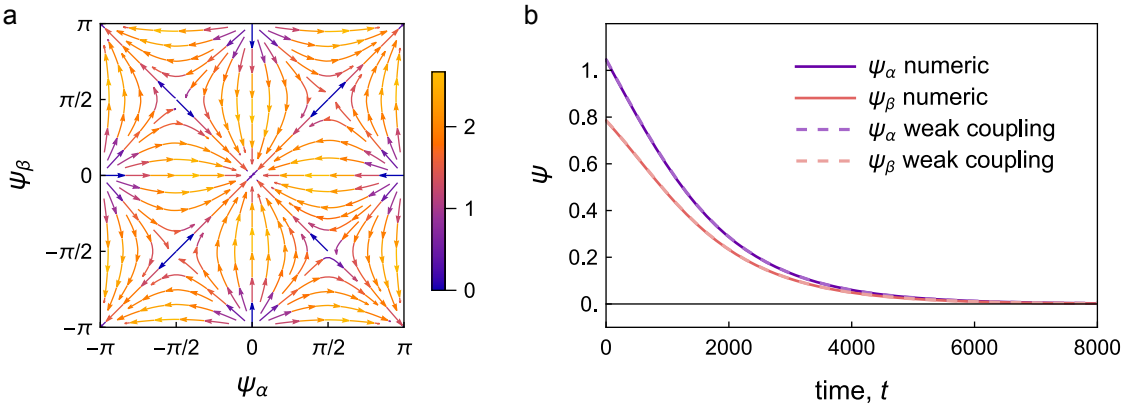

Figure S11: (a) Phase portrait for equations (6.67) showing the angle dynamics for two in-phase oscillators with $\Delta\phi = 0$. Colormap shows the flow velocity for $d = 1$. (b) The angles ψ_α and ψ_β relax to zero at a characteristic rate of $2.68d^{-3} + \mathcal{O}(d^{-4})$. The dashed curve shows the prediction of equation (6.67) for two weakly coupled oscillators separated by a distance $d = 15$ with initial angles $\psi_\alpha(0) = \pi/3$ and $\psi_\alpha(0) = \pi/4$; the solid curve shows the full numerical solution.

Supplementary Note 7: Additional results of swarmalator simulations

7.1 Finite-sized circular clusters

To test the robustness of the circular, oscillatory state observed experimentally and in SD simulations, we performed swarmalator simulations in finite crystalline clusters. Particles were initially placed on a triangular lattice confined within a disk of radius R (cluster size N set by R). The lattice site at the geometric center was removed to mimic the motionless core particle(s) pinned by electroosmotic attraction and geometric confinement. The simulations qualitatively reproduce the experimental and SD trends: as N increases, the mean rotation frequency ω rises and then saturates for large clusters (Fig. S12b), while the circular order remains high (Fig. S12c). For sufficiently large N, however, the circular synchronematic state destabilizes: the circular order parameter S_c drops sharply, signaling the onset of defects and loss of global circular synchrony. These behaviors mirror the experimental and SD observations across cluster sizes.

7.2 Bulk systems with periodic boundaries

To assess the role of geometrical disorder, we performed additional simulations in which the same model was arranged on a perfectly ordered square lattice (Fig. S13). Compared to the randomly initialized configurations, the square lattice displays a higher degree of organization, reflected in larger average nematic and synchronization order parameters.

At vanishing noise amplitudes ($D_\chi \to 0$), the system often becomes trapped in metastable states containing one or two stationary vortices, which restrict the global correlation length. As the noise amplitude increases, fluctuations destabilize these ordered domains and the correlations decay exponentially, indicating a transition to a disordered phase. When the correlation length approaches the system size, finite-size effects can produce apparent long-range order, accounting for the large run-to-run fluctuations observed in S and R at small but finite diffusion coefficients (Fig. S13a–d). Increasing the system size to $N = 8100$ particles reveals a robust proliferation of topological defects, which can occur either in phase or in alignment (Fig. S13e).

In disordered geometries, when non-reciprocal interactions are suppressed, quasi–long-range order characteristic of the XY model is recovered (Fig. S13f).

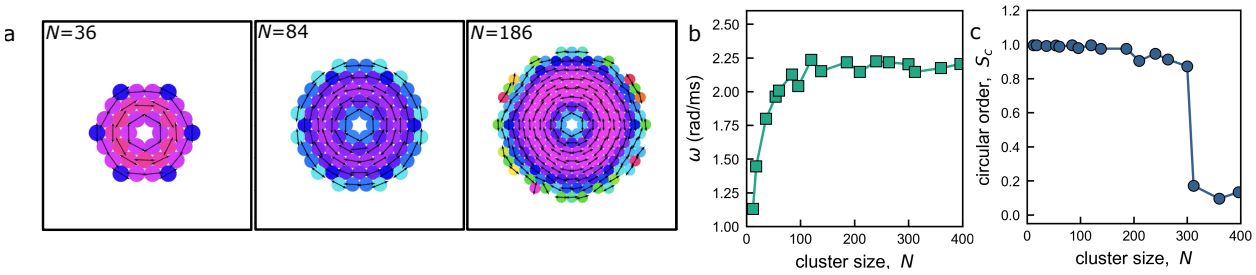

Figure S12: Synchronematic order in circular crystals. (a) Snapshots of simulated clusters with $N = 36$, 84, 186 showing global phase locking with circular alignment (colors denote phase; arrows indicate orientation). The central lattice site is removed to emulate the pinned core. (b) Mean oscillation frequency ω versus cluster size N: ω increases with N and saturates for large clusters. (c) Circular order parameter S_c versus cluster size N: high circular order for small–intermediate clusters, followed by a sharp decline at large N, indicating loss of circular synchronematic order.

7.3 Metachronal waves in 1D oscillator arrays

Previous results on models reminiscent of the Sakaguchi–Kuramoto model predict the existence of one-dimensional metachronal waves stabilized by phase-locked states[16]. Since our model bears similarities to simplified mathematical descriptions of oscillatory beating cilia, we simulated one-dimensional oscillator chains of increasing length under periodic boundary conditions to test whether such metachronal waves can also emerge in this model. Figure S14 shows the resulting space–time representations of the phase field. For small systems ($N = 50$), all oscillators remain phase locked and no spatial modulation develops. When the number of oscillators increases to $N = 200$, a finite phase gradient emerges, indicating the onset of a metachronal wave characterized by a constant phase lag between neighboring oscillators. For still larger chains ($N = 600$), the system supports multiple metachronal domains that propagate in opposite directions, separated by stationary phase defects. These results demonstrate that while small systems synchronize globally, larger periodic chains spontaneously break spatial symmetry and develop traveling metachronal modes whose number and wavelength depend on the system size.

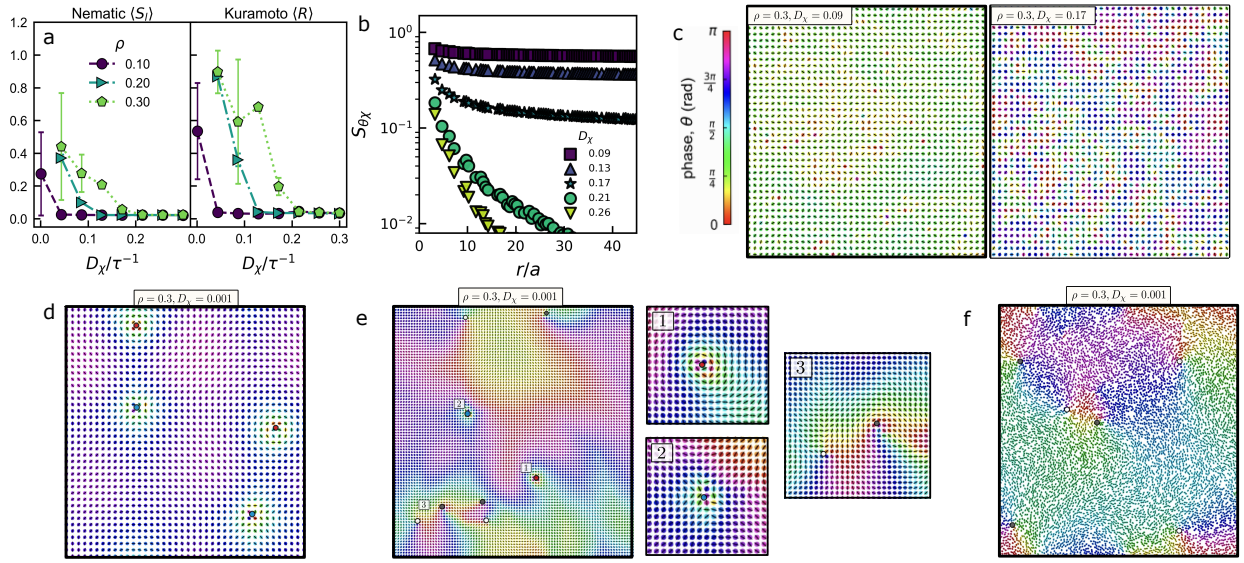

Figure S13: Swarmalator simulations using the interactions derived from weakly coupled oscillator theory. Simulations (a-d) correspond to a square lattice of size 40x40, hence $N = 1600$ particles, and area fraction $\rho = 0.3$. In all cases phases and angles are initialized randomly. In (e-f), we perform additional simulations with $N = 8100$ with density $\rho = 0.3$. In these cases the radii of the particles in the image is doubled to allow a better visualization of the phase. (a) Nematic and Kuramoto order parameters, showing the overall alignment and synchronicity of the lattice. We have performed 5 simulations for each point, and the error-bars represent the variance between the simulations. (b) Synchronematic correlation along the square lattice. For large diffusion coefficients, correlations decay exponentially, while for small diffusion coefficients, quasi-long-range order is observed. (c) Snapshot of the simulations showing small noise correlations in the quasi-long-range order regime (left) and exponential decay regime (right). (d) Lattice simulation for $N = 1600$ particles, $\rho = 0.3$ and $D_\chi = 0.001$ showing alignment defects with opposite (blue and red) charges. (e) Lattice simulation for $N = 8100$ particles, showing alignment defects with opposite (blue and red) charges, and phase defects with opposite charges (white and gray circles). (f) Random geometry simulations of the reduced-order synchronematic model suppressing the even terms in phase, for $N = 8100$ particles, $\rho = 0.3$ and $D_\chi = 0.001$. In the absence of reciprocal interactions, the results are reminiscent of the XY phase model.

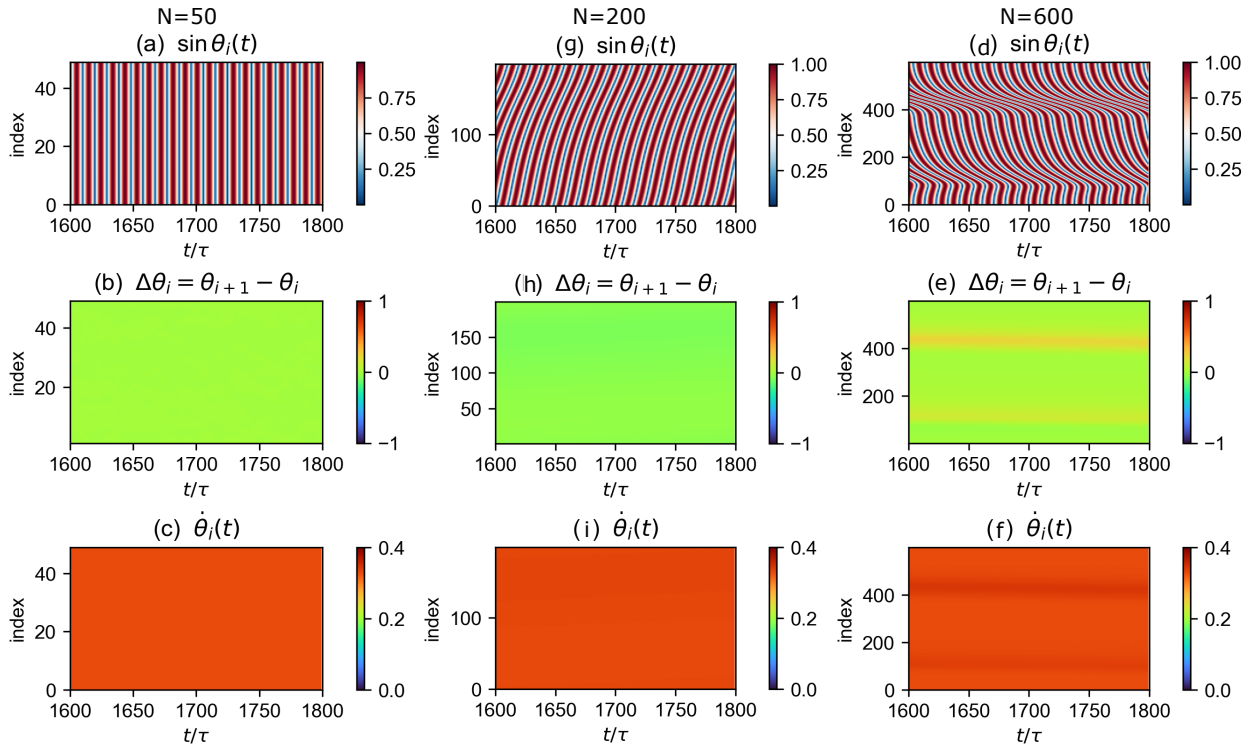

Figure S14: System-size dependence of the metachronal dynamics. Space–time diagrams of the oscillator phase field for systems of different sizes: $N = 50$ (left), $N = 200$ (center), and $N = 600$ (right). (a,g,d) Temporal evolution of $\sin\theta_i(t)$ (color scale) reveals the spatiotemporal organization of the oscillators. (b,h,e) Local phase difference between neighboring oscillators, $\Delta\theta_i = \theta_{i+1} - \theta_i$. (c,i,f) Instantaneous phase velocity, $\dot{\theta}_i(t)$. For small systems ($N = 50$), all oscillators synchronize in phase ($\Delta\theta_i \simeq 0$), and the pattern remains stationary. Increasing the number of oscillators ($N = 200$) allows the emergence of a uniform phase gradient, corresponding to a propagating metachronal wave with a constant phase lag between neighbors. For larger systems ($N = 600$), multiple metachronal domains coexist, separated by narrow transition regions, consistent with the formation of long-wavelength metachronal patterns under periodic boundary conditions. In all cases, the phase velocities converge to a uniform mean value after a short transient, confirming frequency entrainment.

Supplementary Note 8: Swarmalator interactions are non-conservative

In the main text, we present a swarmalator model using the following hydrodynamic interaction functions for the oscillator angles and phases

$$H^{\theta}_{3\mathrm{h}}(\psi_\alpha,\psi_\beta,\Delta\theta) =(A^{\theta}_1\cos\Delta\theta + B^{\theta}_1\sin\Delta\theta + A^{\theta}_3\cos 3\Delta\theta + B^{\theta}_3\sin 3\Delta\theta)\cos\psi_\alpha\cos\psi_\beta \tag{8.1}$$
$$H^{\chi}_{3\mathrm{h}}(\psi_\alpha,\psi_\beta,\Delta\theta) =(A^{\chi}_1\cos\Delta\theta + B^{\chi}_1\sin\Delta\theta + A^{\chi}_3\cos 3\Delta\theta + B^{\chi}_3\sin 3\Delta\theta)\sin\psi_\alpha\cos\psi_\beta \tag{8.2}$$

where $\psi_\alpha = \chi_\alpha - \mu$ and $\psi_\beta = \chi_\beta - \mu$ are the oscillator orientations measured from the line of centers, and $\Delta\theta = \theta_\beta - \theta_\alpha$ is the phase difference. These interaction functions can be interpreted as generalized forces that drive overdamped dynamics of the form

$$\begin{aligned}\dot\phi_\alpha &= F^{\theta}_\alpha = H^{\theta}_{3\mathrm{h}}(\psi_\alpha,\psi_\beta,\Delta\theta)\\ \dot\chi_\alpha &= F^{\chi}_\alpha = H^{\chi}_{3\mathrm{h}}(\psi_\alpha,\psi_\beta,\Delta\theta)\end{aligned}\quad\text{and}\quad\begin{aligned}\dot\phi_\beta &= F^{\theta}_\beta = H^{\theta}_{3\mathrm{h}}(\psi_\beta,\psi_\alpha,-\Delta\theta)\\ \dot\chi_\beta &= F^{\chi}_\beta = H^{\chi}_{3\mathrm{h}}(\psi_\beta,\psi_\alpha,-\Delta\theta)\end{aligned} \tag{8.3}$$

where $\phi = \theta - \omega t$ is the relative phase.

The pairwise forces $\mathbf{F}_{\mathrm{pair}} = [F^{\theta}_\alpha, F^{\chi}_\alpha, F^{\theta}_\beta, F^{\chi}_\beta]^\top$ are **non-conservative**, meaning they cannot be written as the gradient of a scalar potential $\Phi_{\mathrm{pair}}(\psi_\alpha,\psi_\beta,\Delta\theta)$. To show this, we perform a Helmholtz decomposition of the force field in the four-dimensional configuration space $(\phi_\alpha,\chi_\alpha,\phi_\beta,\chi_\beta)$:

$$\mathbf{F}_{\mathrm{pair}} = -\nabla\Phi_{\mathrm{pair}} + \mathbf{F}^{\mathrm{nc}}_{\mathrm{pair}} \quad\text{with}\quad \nabla\cdot\mathbf{F}^{\mathrm{nc}}_{\mathrm{pair}} = 0 \tag{8.4}$$

where $\nabla = [\partial_{\phi_\alpha},\partial_{\chi_\alpha},\partial_{\phi_\beta},\partial_{\chi_\beta}]^\top = [-\partial_{\Delta\theta},\partial_{\psi_\alpha},\partial_{\Delta\theta},\partial_{\psi_\beta}]^\top$ is the gradient with respect to the pair coordinates. By construction, the non-conservative part $\mathbf{F}^{\mathrm{nc}}_{\mathrm{pair}}$ is divergence-free, and the conservative part is curl-free in the four-dimensional sense:

$$(\nabla\times(-\nabla\Phi_{\mathrm{pair}}))_{ij} = \partial_i(\partial_j\Phi_{\mathrm{pair}}) - \partial_j(\partial_i\Phi_{\mathrm{pair}}) = 0 \tag{8.5}$$

so that every mixed derivative of the potential is symmetric. Here, a nonzero four-dimensional curl $(\nabla\times\mathbf{F}_{\mathrm{pair}})_{ij}\neq 0$ indicates that the force field cannot be derived solely from a scalar potential. The unique pair potential characterizing the conservative interactions is

$$\Phi_{\mathrm{pair}}(\psi_\alpha,\psi_\beta,\Delta\theta) = \Big(\tfrac{1}{2}\big(A^{\chi}_1 - B^{\theta}_1\big)\cos\Delta\theta + \tfrac{1}{10}\big(A^{\chi}_3 - 3B^{\theta}_3\big)\cos 3\Delta\theta\Big)\cos\psi_\alpha\psi_\beta \tag{8.6}$$

From this result, one may derive the conservative and non-conservative forces using Eq. (8.4). Below, we examine separately the phase interactions (for fixed orientations) and the orientation interactions (for fixed phases) to highlight conservative and non-conservative features in the decoupled dynamics.

The pairwise interactions are also **non-reciprocal**, meaning that the action of oscillator β on α is not equal and opposite to that of α on β:

$$\mathbf{F}_\alpha \neq -\mathbf{F}_\beta \tag{8.7}$$

where $\mathbf{F}_\alpha = [F^{\theta}_\alpha, F^{\chi}_\alpha]^\top$ and $\mathbf{F}_\beta = [F^{\theta}_\beta, F^{\chi}_\beta]^\top$. Non-reciprocity reflects a breakdown of action–reaction symmetry and arises here because the hydrodynamic flow induced by one oscillator depends on its instantaneous phase and orientation, which generally differ from those of its partner. Reciprocity and conservativity are distinct: reciprocity concerns symmetry under exchange of particles, whereas conservativity asks whether the force field derives from a scalar potential in configuration space.

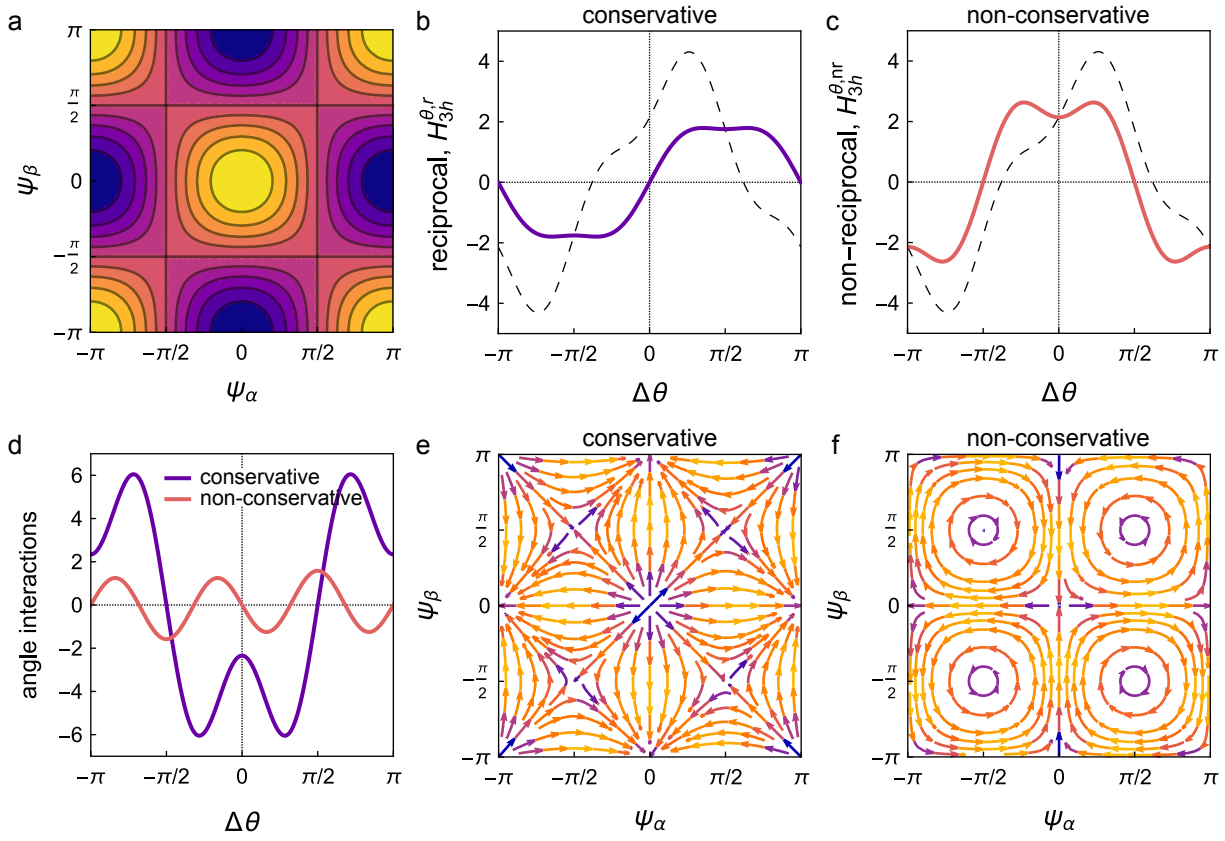

Figure S15: Decomposition of the interaction functions into conservative and non-conservative contributions. (a-c) The phase interaction function $H^{\theta}_{3\mathrm{h}}(\psi_\alpha, \psi_\beta, \Delta\theta)$ of Eq. (8.1) is the product of an angle-dependent prefactor $\cos\psi_\alpha \cos\psi_\beta$ (a) with the sum of reciprocal sine (b) and non-reciprocal cosine (c) functions of the phase difference $\Delta\theta$. (d-f) The angle interaction function $H^{\chi}_{3\mathrm{h}}(\psi_\alpha, \psi_\beta, \Delta\theta)$ of Eq. (8.2) is the product of a phase-dependent prefactor (d) with the associated conservative (e) and non-conservative (f) flow field in the $(\chi_\alpha, \chi_\beta)$ orientation space.

For identical oscillators, the conservative pair potential $\Phi_{\mathrm{pair}}(\psi_\alpha, \psi_\beta, \Delta\theta)$ is, by construction and necessity, invariant under index permutation

$$\Phi_{\mathrm{pair}}(\psi_\alpha, \psi_\beta, \Delta\theta) = \Phi_{\mathrm{pair}}(\psi_\beta, \psi_\alpha, -\Delta\theta) \tag{8.8}$$

Nevertheless, the resulting interactions—particularly, the generalized torques that act to rotate oscillator orientations—are generally non-reciprocal unless the potential depends only on the orientation difference $\psi_\beta - \psi_\alpha$. Because the pair potential contains terms like $\cos\psi_\alpha \cos\psi_\beta$, which mix angle sums and differences, the torques experienced by each oscillator are not equal and opposite even though the underlying potential remains symmetric.

8.1 Phase Interactions

As discussed in the main text and illustrated in Figure 5b, the phase interactions can be decomposed into reciprocal and non-reciprocal contributions as

$$H^{\theta}_{3\mathrm{h}}(\psi_\alpha, \psi_\beta, \Delta\theta) = H^{\theta,\mathrm{r}}_{3\mathrm{h}}(\psi_\alpha, \psi_\beta, \Delta\theta) + H^{\theta,\mathrm{nr}}_{3\mathrm{h}}(\psi_\alpha, \psi_\beta, \Delta\theta) \tag{8.9}$$

where the respective components are defined as

$$\begin{aligned} H_{3\mathrm{h}}^{\theta,\mathrm{r}}(\psi_\alpha,\psi_\beta,\Delta\theta) &= (B_1^\theta \sin\Delta\theta + B_3^\theta \sin 3\Delta\theta)\cos\psi_\alpha\cos\psi_\beta \\ H_{3\mathrm{h}}^{\theta,\mathrm{nr}}(\psi_\alpha,\psi_\beta,\Delta\theta) &= (A_1^\theta \cos\Delta\theta + A_3^\theta \cos 3\Delta\theta)\cos\psi_\alpha\cos\psi_\beta \end{aligned} \tag{8.10}$$

The reciprocal and non-reciprocal contributions are, respectively, odd and even under permutation of the indices α and β. Figure S15a-c illustrates this decomposition as the product of an angle-dependent prefactor $\cos\psi_\alpha\cos\psi_\beta$ (Fig. S15a) with the sum of reciprocal sine (Fig. S15b) and non-reciprocal cosine (Fig. S15c) functions of the phase difference.

The reciprocal contribution, $H_{3\mathrm{h}}^{\theta,\mathrm{r}}(\psi_\alpha,\psi_\beta,\Delta\theta) = -\frac{1}{2}G_3^\theta(\psi_\alpha,\psi_\beta,\Delta\theta)$, governs the evolution of the phase difference $\Delta\theta$ and describes the tendency of oscillator pairs to synchronize (Fig. S10). The non-reciprocal contribution, $H_{3\mathrm{h}}^{\theta,\mathrm{nr}}(\psi_\alpha,\psi_\beta,\Delta\theta) = \frac{1}{2}S_3^\theta(\psi_\alpha,\psi_\beta,\Delta\theta)$, instead governs the evolution of the mean phase and captures the tendency of oscillator pairs to oscillate faster when synchronized. The reciprocal phase interactions are conservative, as they derive from the scalar potential

$$\Phi_{\mathrm{pair}}^\theta(\psi_\alpha,\psi_\beta,\Delta\theta) = -(B_1^\theta\cos\Delta\theta + B_3^\theta\cos 3\Delta\theta)\cos\psi_\alpha\cos\psi_\beta \tag{8.11}$$

In this case, the correspondence between reciprocity and conservativity is exact: the reciprocal interactions correspond to the conservative component of the force field, and the non-reciprocal interactions correspond to the non-conservative component.

8.2 Angle Interactions

The angle interactions can be decomposed into conservative and non-conservative contributions in the two-dimensional orientation space (χ_α,χ_β):

$$H_{3\mathrm{h}}^{\chi}(\psi_\alpha,\psi_\beta,\Delta\theta) = H_{3\mathrm{h}}^{\chi,\mathrm{c}}(\psi_\alpha,\psi_\beta,\Delta\theta) + H_{3\mathrm{h}}^{\chi,\mathrm{nc}}(\psi_\alpha,\psi_\beta,\Delta\theta) \tag{8.12}$$

where the respective contributions are defined as

$$\begin{aligned} H_{3\mathrm{h}}^{\chi,\mathrm{c}}(\psi_\alpha,\psi_\beta,\Delta\theta) &= (A_1^\chi \cos\Delta\theta + A_3^\chi \cos 3\Delta\theta)\sin\psi_\alpha\cos\psi_\beta \\ H_{3\mathrm{h}}^{\chi,\mathrm{nc}}(\psi_\alpha,\psi_\beta,\Delta\theta) &= (B_1^\chi \sin\Delta\theta + B_3^\chi \sin 3\Delta\theta)\sin\psi_\alpha\cos\psi_\beta \end{aligned} \tag{8.13}$$

Figure S15d-f illustrates this decomposition as the product of a phase-dependent prefactor (Fig. S15d) with the associated flow field in the (χ_α,χ_β) orientation space (Fig. S15e,f). The conservative component corresponds to gradients of the pair potential

$$\Phi_{\mathrm{pair}}^\chi(\psi_\alpha,\psi_\beta,\Delta\theta) = -(A_1^\chi\cos\Delta\theta + A_3^\chi\cos 3\Delta\theta)\cos\psi_\alpha\cos\psi_\beta \tag{8.14}$$

while the non-conservative component gives rise to circulating flows that cannot be expressed as the gradient of a potential (Fig. S15f).

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